

ELASTIC

Documentation

文档

中英对照 · 14 篇文章
面向 Java 开发

官方文档快照：2026-09-26
frc-elastic.gitbook.io/docs

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GIF 在 PDF 中显示静态预览，并附原动画链接。

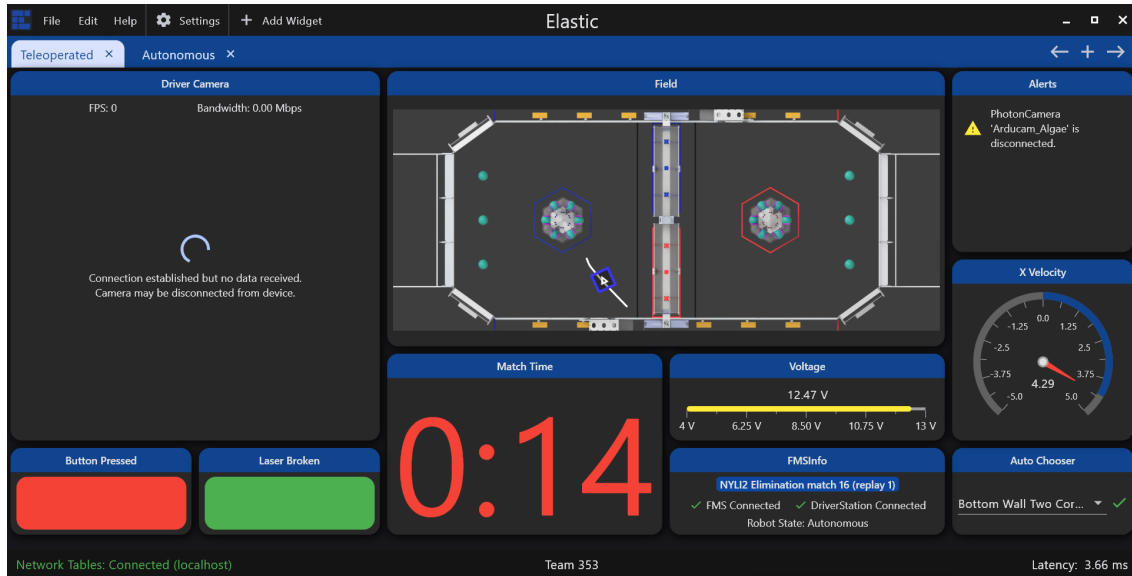
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Home / 首页

Elastic is a simple and modern dashboard for FRC. It was built with the goal of being a drag & drop dashboard with an easy setup and elegant UI designed for a high-pressure competition environment.

Elastic 是一款简洁、现代的 FRC Dashboard。它的开发目标是提供可拖放操作、易于设置且界面简洁的 Dashboard，适用于压力较大的比赛环境。



Getting Started / 入门

- App Navigation
应用导航
- Connecting to your Robot
连接到机器人

Creating and Customizing your Dashboard / 创建和定制 Dashboard

- Adding and Customizing Widgets
添加和定制控件
- Adding and Customizing Layouts
添加和定制布局

Features & References / 功能与参考资料

- Widgets List & Properties Reference
控件列表与属性参考
- Shuffleboard API Integration
Shuffleboard API 集成
- Robot Notifications
机器人通知

Installation / 安装

Elastic is supported on the following Operating Systems

Elastic 支持以下操作系统

Operating System 操作系统	Supported 是否支持
Windows x86_64	[Yes]
Windows aarch64	[No]
macOS x86_64	[Yes]
macOS arm64	[Yes]
Linux x86_64	[Yes]
Linux aarch64	[No]

To install Elastic, head to the [Releases](#) page and download the file for your respective operating system.

要安装 Elastic，请前往 [Releases](#) 页面，下载对应操作系统的文件。

App Navigation / 应用导航

When the app is first opened, there will be no widgets on the screen, just 2 empty tabs

首次打开应用时，屏幕上没有控件，只有 2 个空选项卡



On Startup

启动时

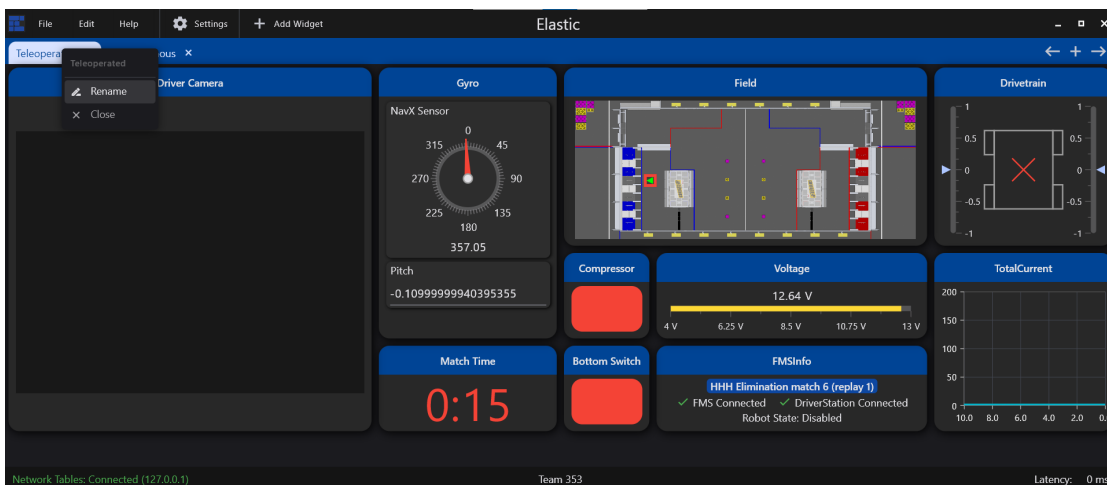
Tab Bar / 选项卡栏

You click on the name of each tab to switch to that tab, or you can use the shortcut `Ctrl + <Tab Number>`. New tabs can be created using the `+` button on the tab bar, and be closed by clicking the `x` button on the tab.

点击各选项卡的名称即可切换到该选项卡，也可以使用快捷键 `Ctrl + <Tab Number>`。使用选项卡栏上的 `+` 按钮可以创建新选项卡，点击选项卡上的 `x` 按钮可以关闭它。

To rename a tab, simply right click on the tab, click rename, and enter the new name of the tab.

要重命名选项卡，只需右键点击该选项卡，点击重命名，然后输入新的选项卡名称。



Renaming a Tab / 重命名选项卡

Saving, Exporting, and Importing / 保存、导出和导入

To save your current layout, at the top of your screen, click **File -> Save**, or use the shortcut **Ctrl + S**. A message will pop up on the screen once the layout is successfully saved.

要保存当前布局，请在屏幕顶部点击 **File -> Save**，或使用快捷键 **Ctrl + S**。布局成功保存后，屏幕上会弹出一条消息。

When you save the layout, it is stored internally. To export the layout as a JSON file, at the top of your screen, click **File -> Export Layout**, or you can use the shortcut **Ctrl + Shift + S**. A file explorer window will pop up, select the location you want to save your file in.

保存布局时，布局存储在应用内部。要将布局导出为 JSON 文件，请在屏幕顶部点击 **File -> Export Layout**，或使用快捷键 **Ctrl + Shift + S**。随后会弹出文件浏览窗口，选择要保存文件的位置。

To import a layout from an existing JSON file, click on **File -> Open Layout**, or use the shortcut **Ctrl + O**. Select the JSON file that you want to use to import. It is strongly recommended to not modify the JSON file manually unless if you know what you are doing, as incorrect formatting can cause the program to fail when loading.

要从现有 JSON 文件导入布局，请点击 **File -> Open Layout**，或使用快捷键 **Ctrl + O**。选择要用于导入的 JSON 文件。强烈建议不要手动修改 JSON 文件，除非你清楚自己在做什么，因为格式错误可能导致程序加载失败。

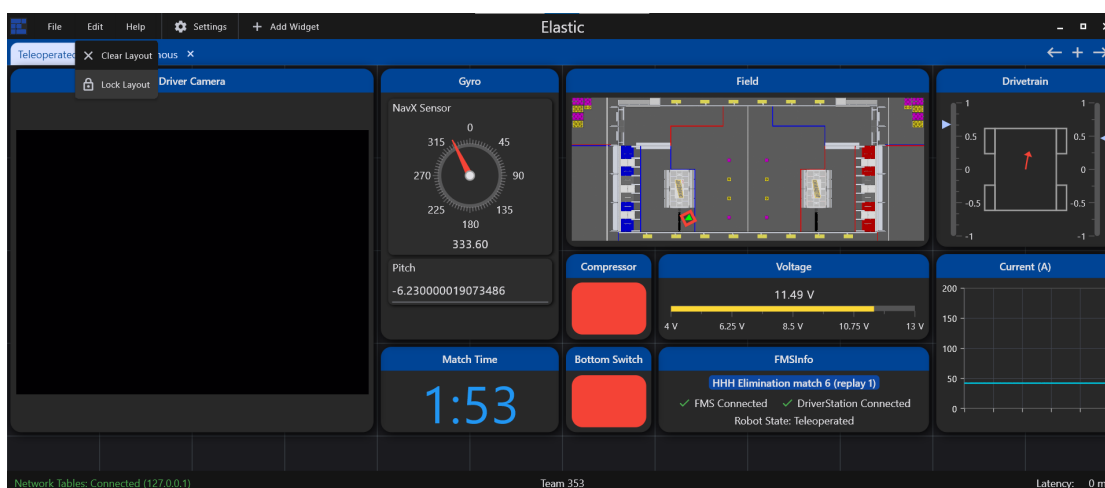
Locking Layout / 锁定布局

Some widgets such as number sliders can be difficult to control as the widget might be dragged when controlling the slider. Sometimes you might want a way of making sure that no widgets are being modified such as during competitions or driver practice. For these situations, you can lock the layout which prevents widgets from being dragged, modified, or removed.

数值滑块等控件可能不易操作，因为在控制滑块时可能会拖动整个控件。有时你可能希望确保任何控件都不会被修改，例如比赛或驾驶练习期间。对于这些情况，可以锁定布局，防止控件被拖动、修改或删除。

To lock the layout, click on **Edit -> Lock Layout**.

要锁定布局，请点击 **Edit -> Lock Layout**。



Locking Layout

锁定布局

When the layout is locked, there will be a lock icon next to the menu bar. When the layout is locked, you will be unable to add, move, modify widgets, or manage tabs. It's important to note that you also cannot add widgets via the Shuffleboard API as well.

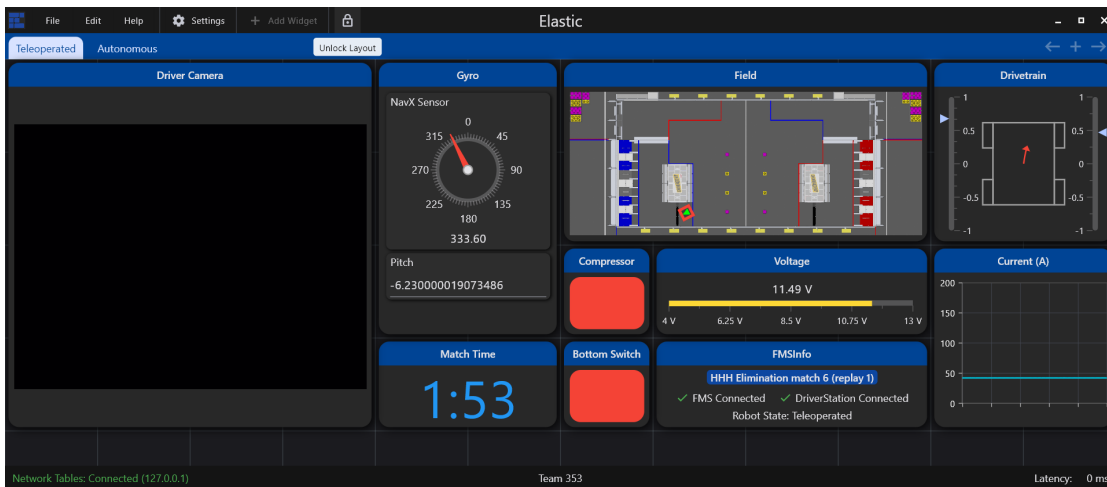
布局锁定后，菜单栏旁边会出现锁形图标。锁定布局时，无法添加、移动或修改控件，也无法管理选项卡。需要注意的是，此时也不能通过 Shuffleboard API 添加控件。



Locked Layout / 已锁定的布局

To unlock the layout, simply click the lock icon at the top of the screen.

要解锁布局，只需点击屏幕顶部的锁形图标。



Unlocking Layout

解锁布局

Connecting To Your Robot / 连接到机器人

In order for the program to connect to Network Tables, you need to be connected to your robot, and the program needs to know what the IP address of the robot is. There are several ways options to do this. If you click the settings gear at the top of the screen, there will be a dropdown menu for IP address connection.

要让程序连接到 NetworkTables，你需要先连接到机器人，并让程序知道机器人的 IP 地址。有几种可选方式。点击屏幕顶部的设置齿轮按钮，会出现用于选择 IP 地址连接方式的下拉菜单。

Driver Station

This is the easiest and recommended way of obtaining the IP address from the robot, as it requires very little manual work. First, connect to your robot, then open the Driver Station. Once the driver station is opened, the program will automatically retrieve the IP address from the TCP Port in the DS.

这是获取机器人 IP 地址最简单且推荐的方式，因为它几乎不需要手动操作。首先连接到机器人，然后打开 Driver Station。Driver Station 打开后，程序会自动通过 DS 中的 TCP 端口获取 IP 地址。

Team Number (10.TE.AM.2) / 队伍编号 (10.TE.AM.2)

NOTE: This will not work when running a robot simulation

说明：运行机器人仿真时，此方式无效

This will connect to network tables using the radio's default IP address for your team number (10.TE.AM.2). For example, if your team number is 353, it will use the IP address 10.3.53.2

此方式使用与队伍编号对应的机器人无线电默认 IP 地址（10.TE.AM.2）连接到 NetworkTables。例如，如果队伍编号为 353，则使用 IP 地址 10.3.53.2

For more information: see [IP Configuration - FIRST Robotics Competition documentation](#)

有关详细信息，请参阅 [IP 配置 - FIRST Robotics Competition 文档](#)

RoboRIO mDNS (roboRIO-###-FRC.local) / RoboRIO mDNS (roboRIO-###-FRC.local)

NOTE: This will not work when running a robot simulation

说明：运行机器人仿真时，此方式无效

This will use the roboRIO mDNS to connect to your robot. If your team number is 353, it will attempt to connect with the mDNS address: roboRIO-353-FRC.local

此方式使用 roboRIO mDNS 连接到机器人。如果队伍编号为 353，则会尝试连接到 mDNS 地址：
roboRIO-353-FRC.local

localhost

NOTE: This will only work when running a robot simulation

说明：此方式仅在运行机器人仿真时有效

This will connect to network tables with the IP address: localhost

此方式使用 IP 地址 localhost 连接到 NetworkTables

Custom / 自定义

This method is generally not recommended, but is available in case if it is needed. When you select the "Custom" option, you will be able to type your IP address in the "IP Address" text box.

通常不推荐此方式，但在需要时可以使用。选择 "Custom" 选项后，可以在 "IP Address" 文本框中输入 IP 地址。

Connection Status / 连接状态

At the bottom of the screen, there is text displaying your connection status to Network Tables. If you are not connected, there will be red text displaying Network Tables: Disconnected. If you are connected, then there will be green text displaying Network Tables: Connected (<IP Address of Connection>).

屏幕底部会显示与 NetworkTables 的连接状态。未连接时，会以红色文字显示 Network Tables: Disconnected；已连接时，会以绿色文字显示 Network Tables: Connected (<IP Address of Connection>)。

When you are disconnected from your robot, all widgets will be disabled.

与机器人断开连接时，所有控件都会被禁用。

Adding and Customizing Widgets / 添加和定制控件

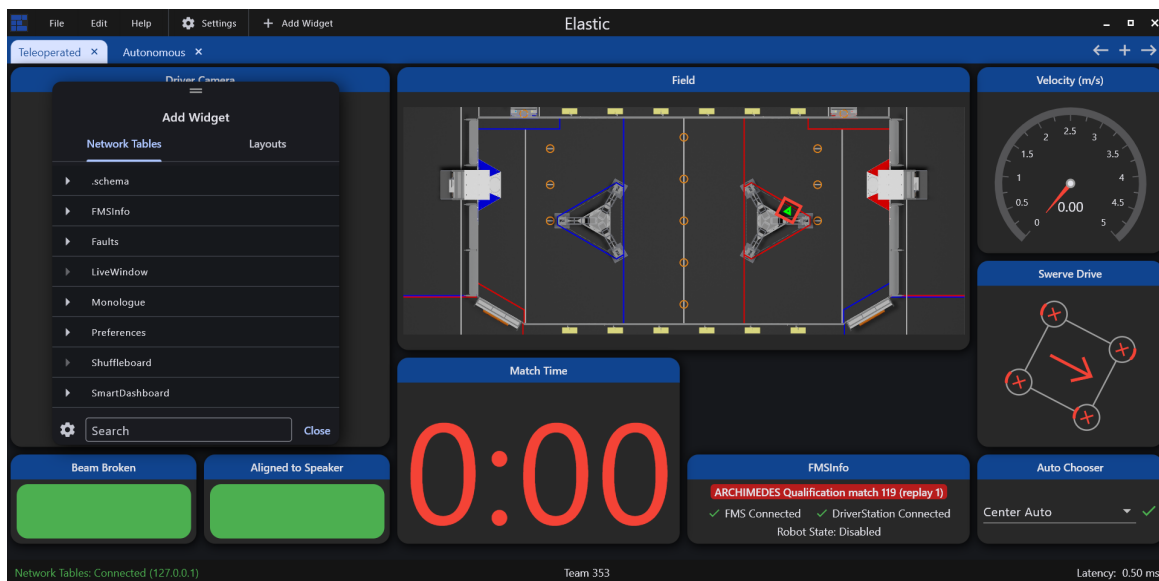
Widgets are the cards on the grid that display information about the robot.

控件是网格中用于显示机器人信息的卡片。

Adding Widgets / 添加控件

On the menu bar at the top of the screen, there is a button that says "Add Widget" with a +. Clicking it will open the widget dialog, which will look like this. You must be connected to your robot in order for tiles to appear.

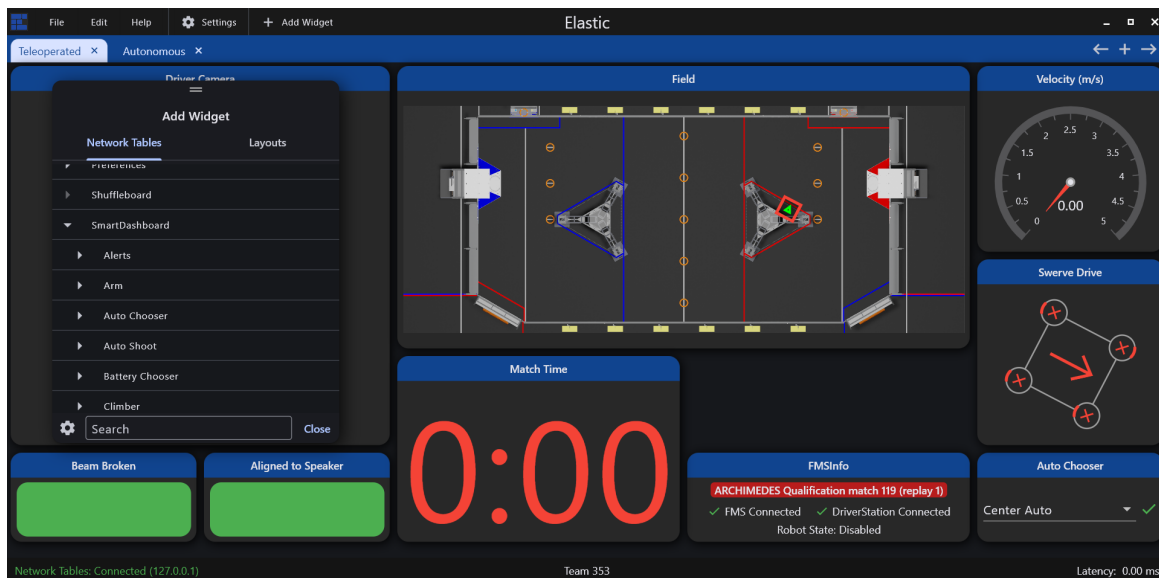
屏幕顶部的菜单栏上有一个带 + 的 "Add Widget" 按钮。点击它会打开控件对话框，如下图所示。必须连接到机器人，对话框中才会出现条目。



Widget Dialog / 控件对话框

By clicking on each table name, you will reveal any sub-tables that Network Tables has.

点击各表的名称，可以展开 NetworkTables 中的子表。



Sub-Tables / 子表

If a topic is a struct type, Elastic will decode the struct and allow you to display individual fields from it. For example, a ChassisSpeeds struct has vx, vy, and omea fields.

如果一个 topic 是 struct 类型, Elastic 会解码该 struct, 并允许单独显示其中的字段。例如, ChassisSpeeds struct 有 vx、vy 和 omea 字段。



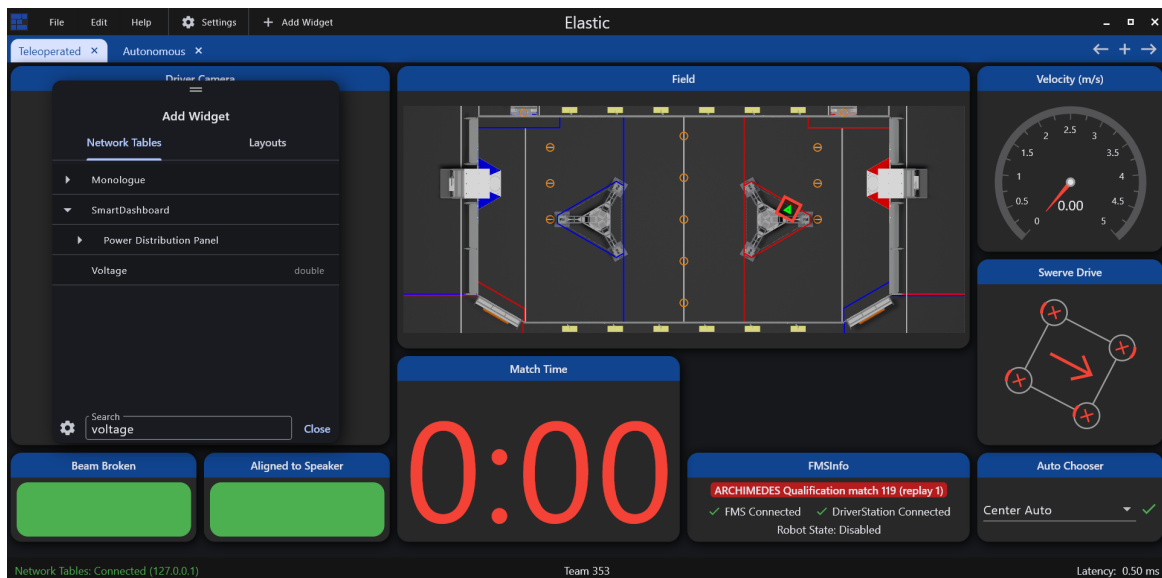
Displaying Struct Fields / 显示 struct 字段

Displaying Struct Fields

显示 struct 字段

To make finding the data you want to display easier, you can type a search query into the search box at the bottom of the dialog. This will only display values that match the search query.

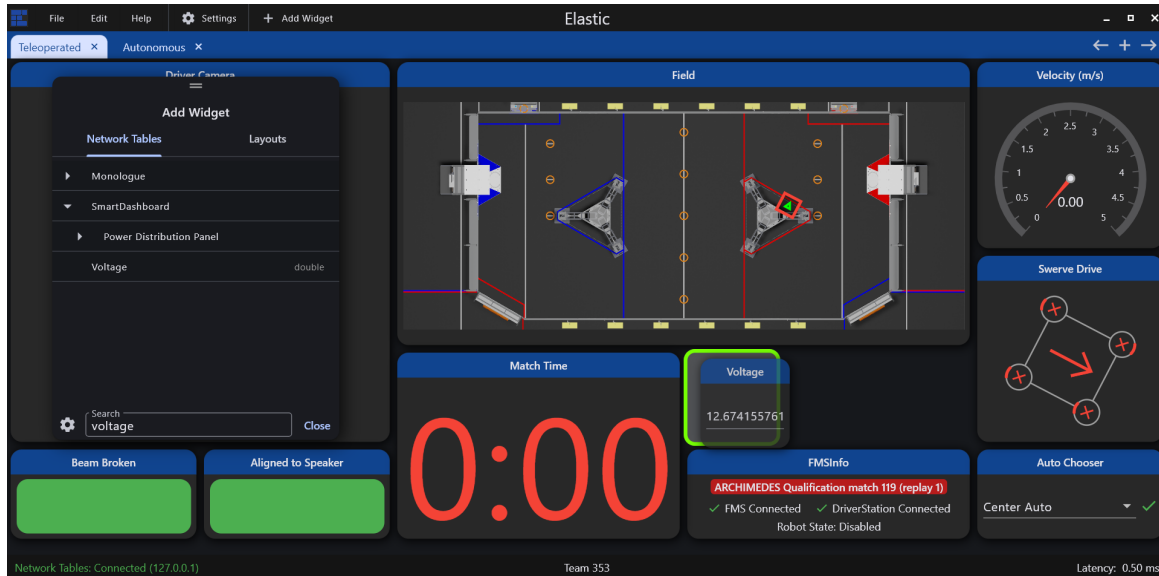
为了更容易找到要显示的数据, 可以在对话框底部的搜索框中输入搜索内容。这样只会显示与搜索内容匹配的值。



Search Query / 搜索内容

When you find the data for the widget you would like to add, simply click on it, and drag it to the location you want to place the widget. A red or green outline will appear, indicating whether or not the location you are hovering on is valid.

找到要添加的控件数据后，只需点击并将其拖到要放置控件的位置。此时会出现红色或绿色轮廓，表示鼠标当前悬停的位置是否有效。



Dragging a Widget to the Grid / 将控件拖到网格中

A widget will be dragged only if there is a valid type for that widget. For example, if you try to drag a boolean value from the dialog, a boolean box will be dragged by your mouse cursor. If you drag a table such as `/Shuffleboard`, which does not have a data type property to it, nothing will appear under your mouse cursor.

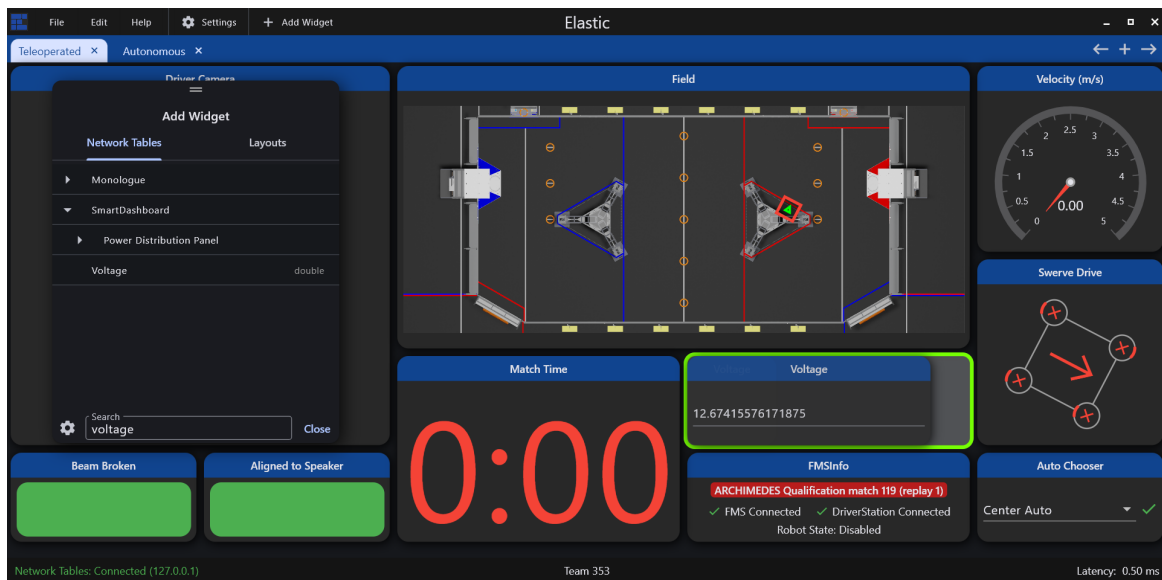
只有存在有效控件类型时，才会拖出控件。例如，尝试从对话框中拖出布尔值时，鼠标光标下会出现一个布尔值方框。如果拖动 `/Shuffleboard` 这样没有数据类型属性的表，鼠标光标下不会出现任何内容。

When you let go of your mouse, the widget will be added to the grid.

松开鼠标时，控件会被添加到网格中。

After you place a widget on the grid, you can resize it by dragging the edges of the widget's box.

将控件放到网格中后，可以通过拖动控件方框的边缘调整其大小。

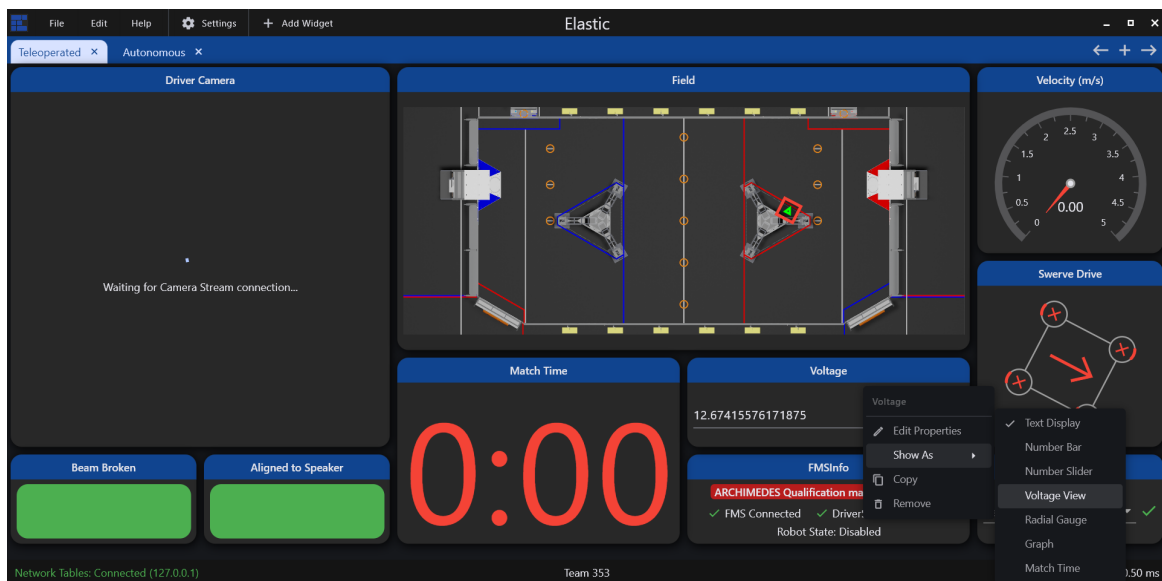


Resizing a Widget / 调整控件大小

Customizing Widgets / 定制控件

When you right click on a widget container (the card that holds the widget), a context menu will appear with several options: Edit Properties, OR Remove.

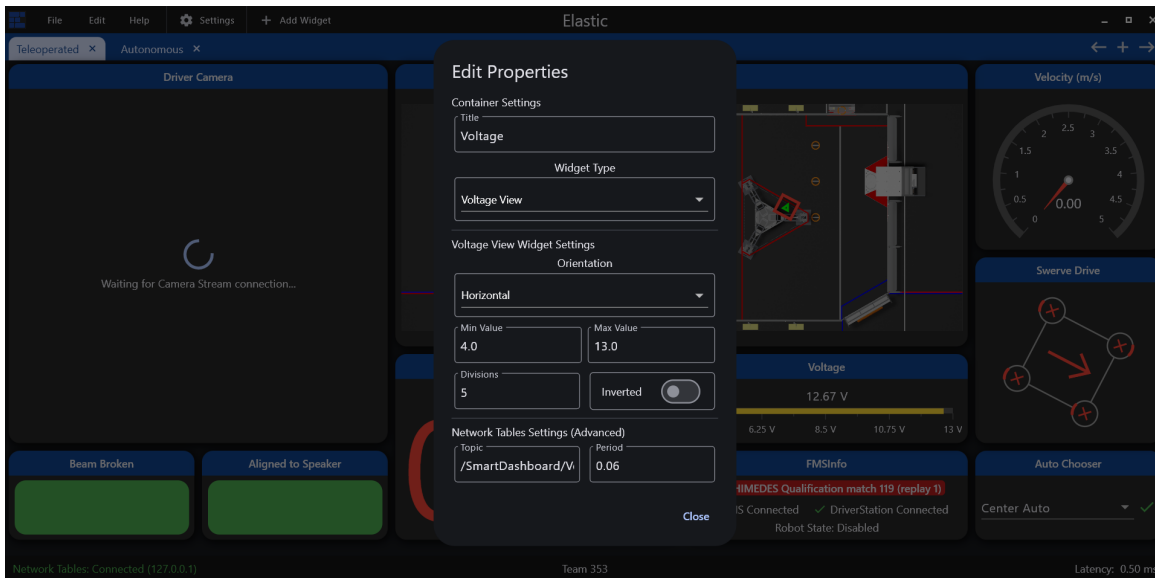
右键点击控件容器（承载控件的卡片）时，会出现一个包含 Edit Properties 和 Remove 选项的上下文菜单。



Right Click Menu / 右键菜单

Clicking Remove will remove the widget from the grid, and clicking Edit Properties will open a menu with several customization options.

点击 Remove 会将控件从网格中移除；点击 Edit Properties 会打开包含多个定制选项的菜单。



Edit Properties Menu / 编辑属性菜单

Every widget will have Container Settings and Network Tables Settings. The container settings allow you to change the title and widget type. Depending on what type of data you are displaying, you are able to change the widget type. You must be connected to your robot for any additional options to appear.

每个控件都有 Container Settings 和 Network Tables Settings。容器设置允许修改标题和控件类型。可选控件类型取决于所显示的数据类型。必须连接到机器人，才会出现其他选项。

Under Container Settings, there will be settings unique for the widget type, titled <Widget Type> Settings. Not every widget will have unique properties to it, if your widget does not have any unique edit properties, there will not be anything listed.

在 Container Settings 下，会显示该控件类型专有的设置，标题为 <Widget Type> Settings。并非每个控件都有专有属性；如果控件没有专有的可编辑属性，此处不会列出任何内容。

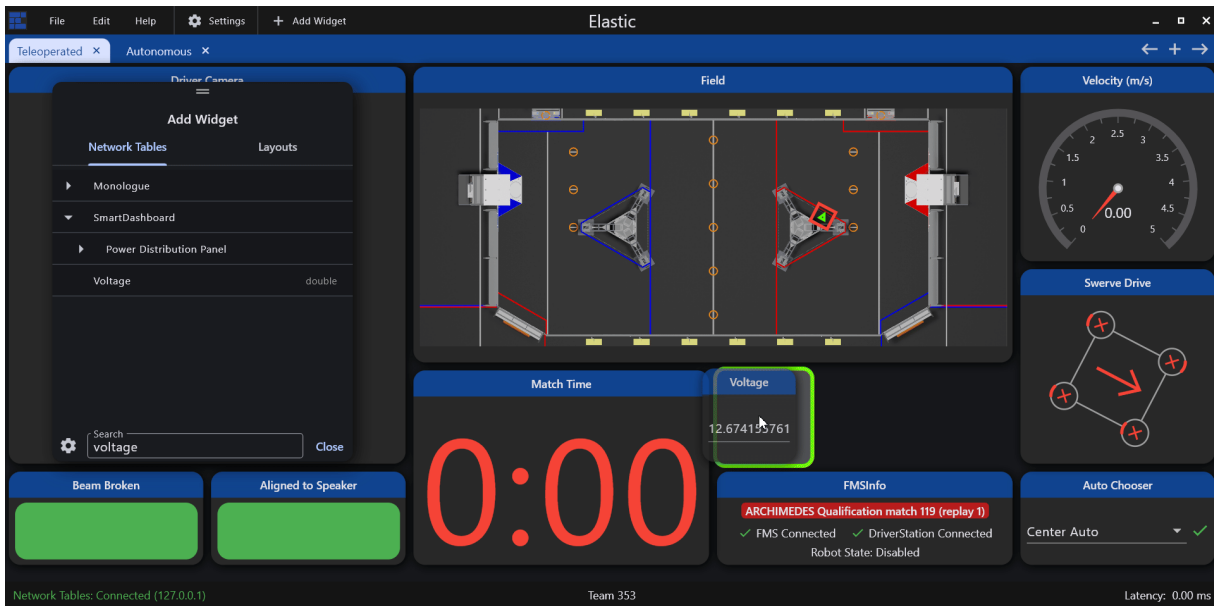
Under Network Tables Settings, there will be several advanced settings allowing you to change the topic that the widget will be receiving data from, along with its update period. It is recommended to not change any of these unless you are familiar with the basics of Network Tables. For more info on Network Tables, see [What is NetworkTables - FIRST Robotics Competition Documentation](#).

在 Network Tables Settings 下，有几个高级设置，可修改控件接收数据的 topic 及其更新周期。除非熟悉 NetworkTables 基础，否则建议不要修改这些设置。有关 NetworkTables 的详细信息，请参阅 [什么是 NetworkTables - FIRST Robotics Competition 文档](#)。

Adding a Voltage Widget (Example) / 添加电压控件（示例）

This gif is an example of how to add a widget. This adds a widget that displays the voltage, and changes a few of its settings.

此 GIF 展示了如何添加控件：添加一个显示电压的控件，并修改其部分设置。



Adding Voltage Example / 添加电压控件示例

Animated GIF: [view original](#) / 动画 GIF: [查看原动画](#)

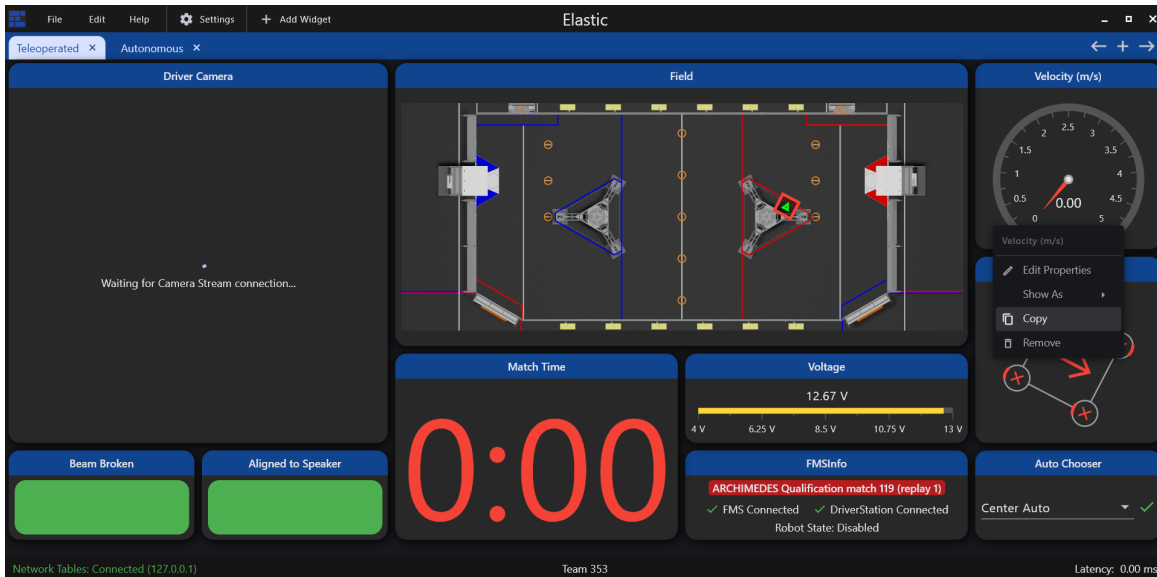
Copying and Pasting Widgets / 复制和粘贴控件

Elastic supports copying and pasting widgets and layouts between different tabs. When right-clicking on a widget, there is an option to copy it to the clipboard.

Elastic 支持在不同选项卡之间复制和粘贴控件及布局。右键点击控件时，可以选择将其复制到剪贴板。

Note: This is separate from the system clipboard

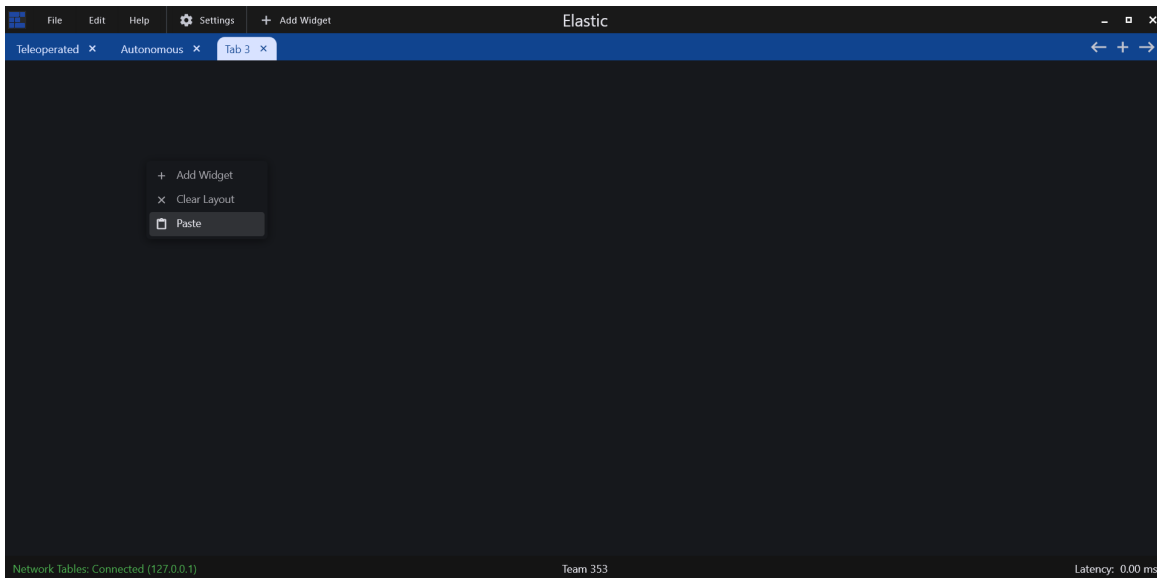
说明：此剪贴板与系统剪贴板相互独立



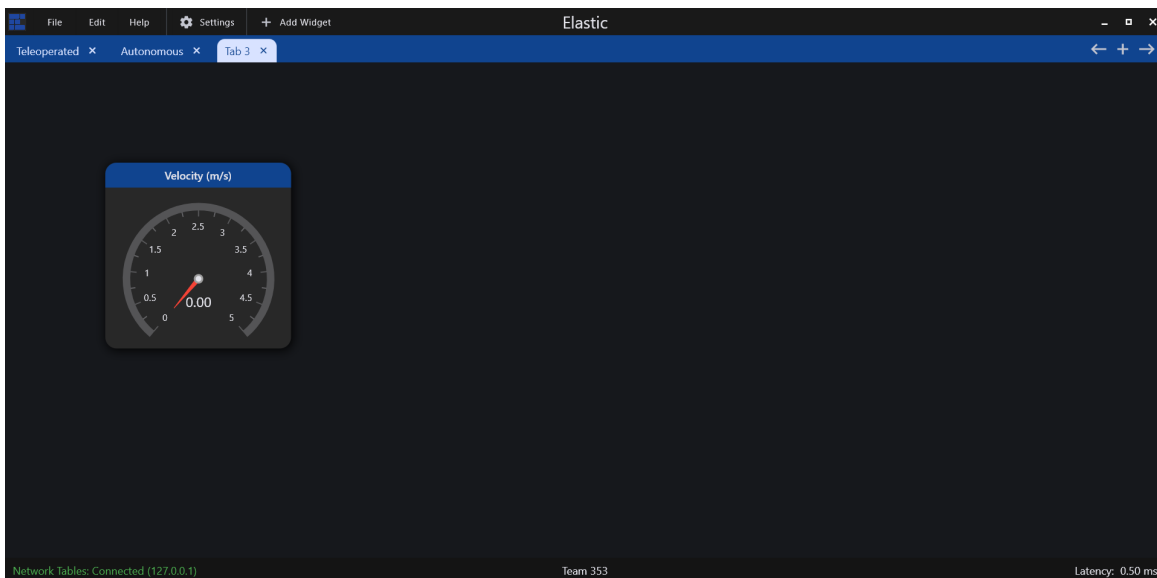
Copying Widget / 复制控件

Right-clicking in an empty spot will show a button to paste the widget. This will attempt to insert the widget into the grid with the top left corner of the widget in the grid space that the cursor is hovering in. If the widget does not fit on the grid, then it will not paste.

右键点击空白位置，会显示粘贴控件的按钮。该操作会尝试将控件插入网格，使控件的左上角位于光标悬停的网格单元内。如果控件无法放入网格，则不会粘贴。



Pasting Widget / 粘贴控件



Widget After Pasting / 粘贴后的控件

Camera Streams / 摄像头视频流

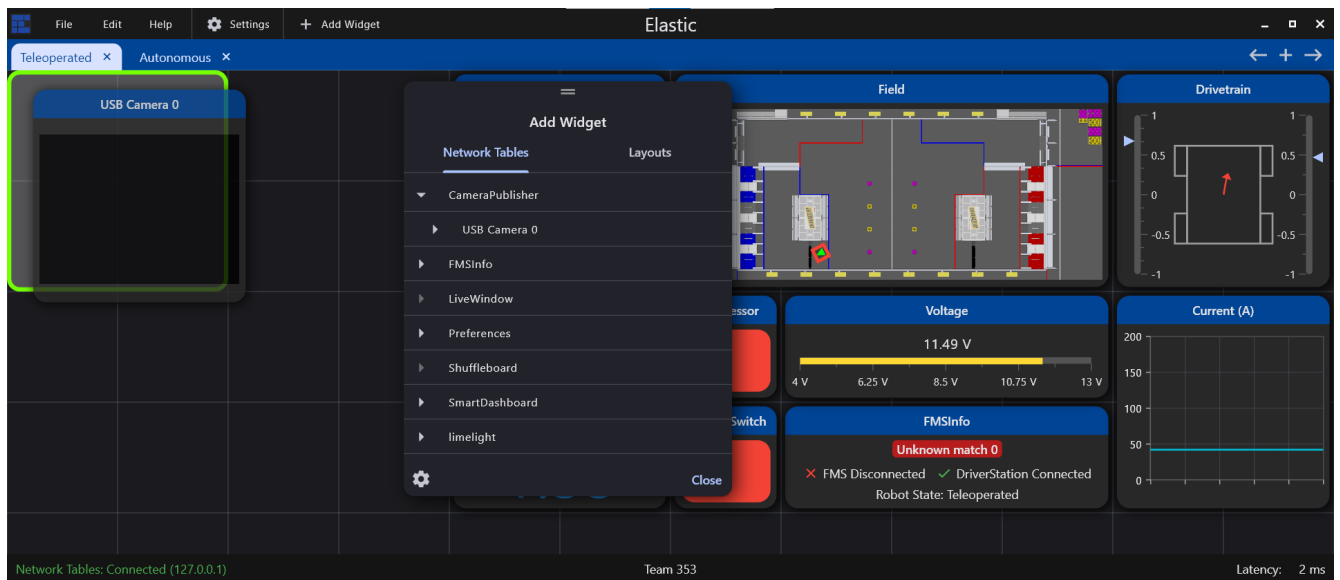
Elastic only supports adding camera streams that are published to the RoboRIO's Camera Server. If you have a USB camera plugged into your RoboRIO, adding this line of code in your `robotInit()` method will allow you to add a camera stream to your dashboard.

Elastic 仅支持添加发布到 RoboRIO 的 Camera Server 的摄像头视频流。如果有 USB 摄像头连接到 RoboRIO，在 `robotInit()` 方法中添加下面这行代码，即可将摄像头视频流添加到 Dashboard。

```
CameraServer.startAutomaticCapture();
```

To add a camera stream, navigate to the `CameraPublisher` table, and there will be different tables for different camera streams that can be viewed. Dragging one of these tables will create a Camera Stream widget that displays live video from the camera.

要添加摄像头视频流，请进入 `CameraPublisher` 表，其中会有对应不同可查看视频流的表。拖动其中一个表，即可创建显示摄像头实时视频的 Camera Stream 控件。



Adding Camera Stream / 添加摄像头视频流

Adding and Customizing Layouts / 添加和定制布局

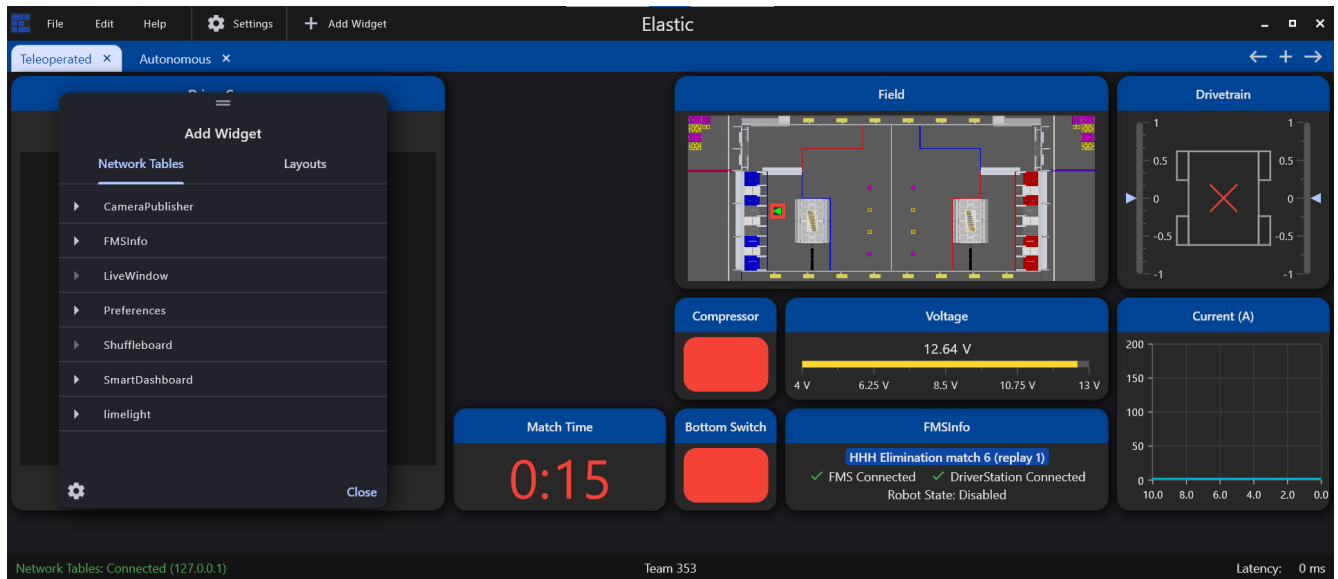
Widgets can be added into layouts that display multiple widgets compacted into one card.

可以将控件添加到布局中，在一张卡片内紧凑地显示多个控件。

Adding a Layout / 添加布局

To add a layout, first click the "Add Widget" button on the top of the screen. Then, click the tab that says "Layouts".

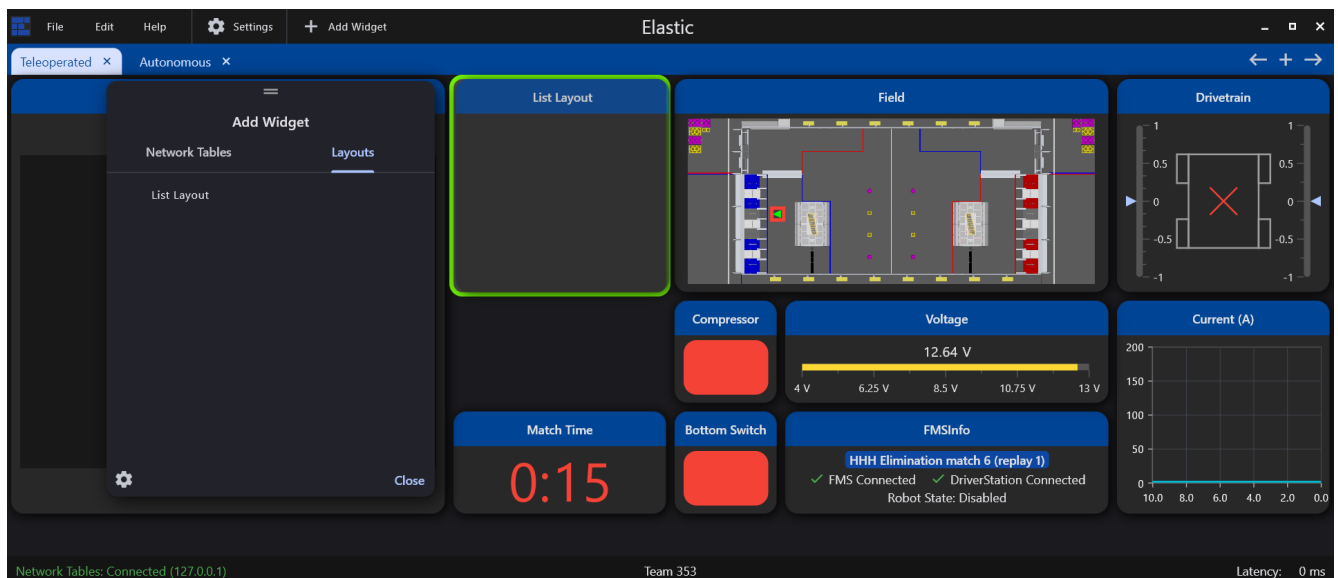
要添加布局，首先点击屏幕顶部的 "Add Widget" 按钮，然后点击 "Layouts" 选项卡。



Layout Tab / 布局选项卡

To drag a layout onto the screen, click the tile for the layout you want to add, and drag it onto the screen.

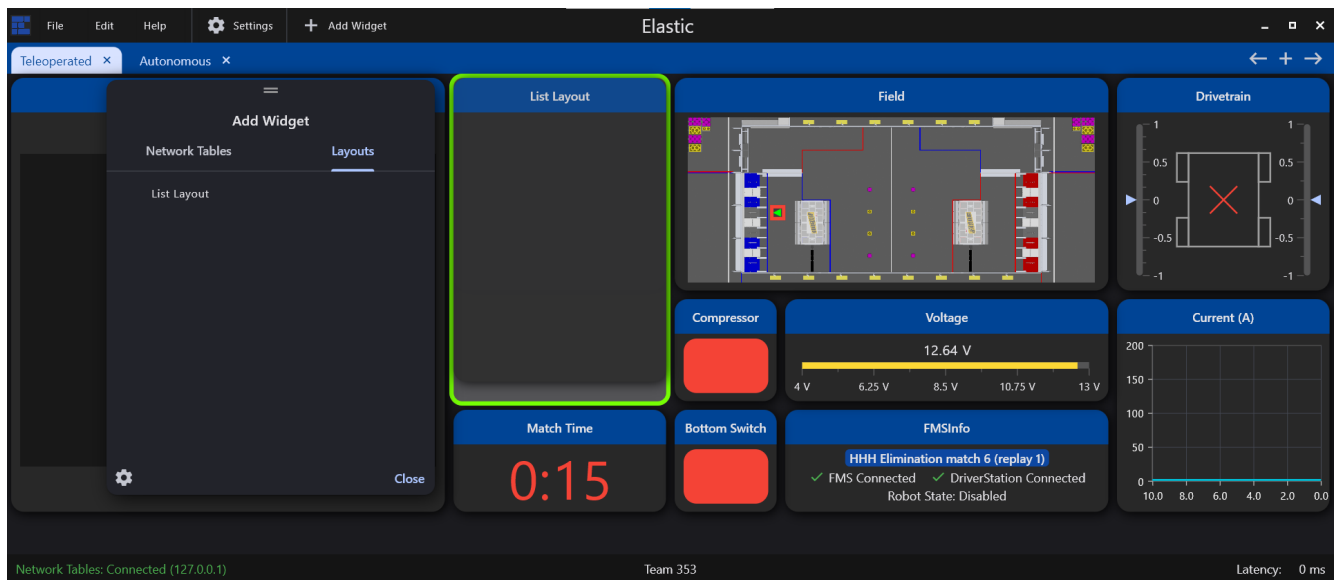
要将布局拖到屏幕上，请点击要添加的布局条目，并将其拖到屏幕上。



Dragging list layout onto the grid / 将列表布局拖到网格中

Just like with regular widgets, a layout can be resized by clicking and dragging the edges of the box.

与普通控件一样，可以点击并拖动方框边缘来调整布局大小。

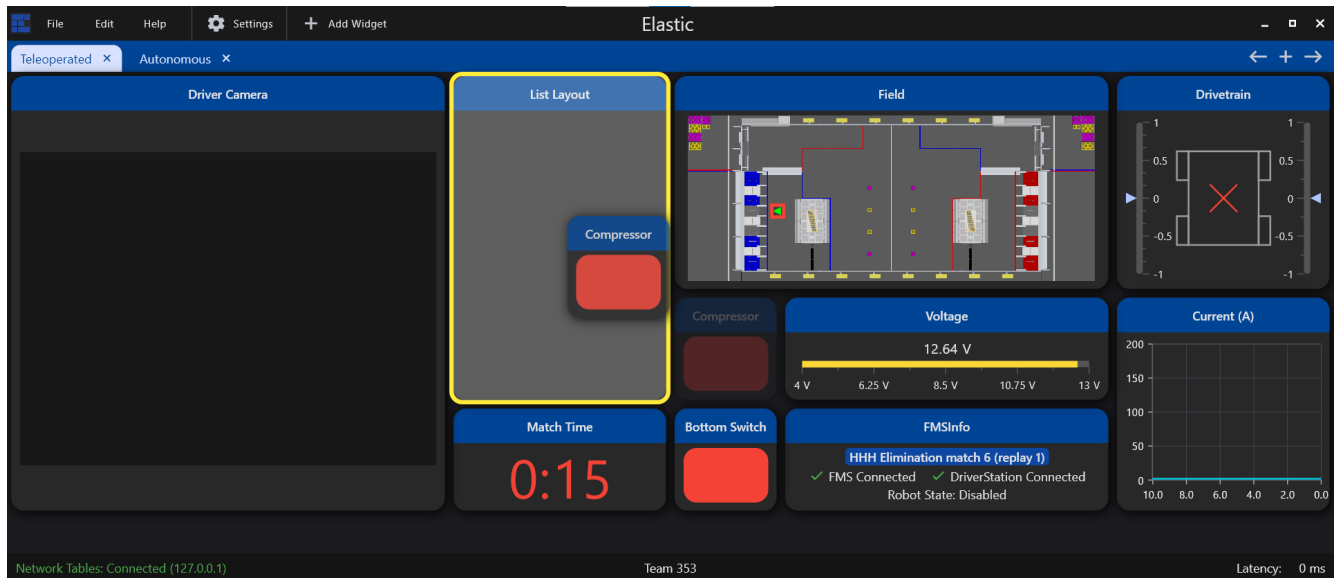


Resizing list layout / 调整列表布局大小

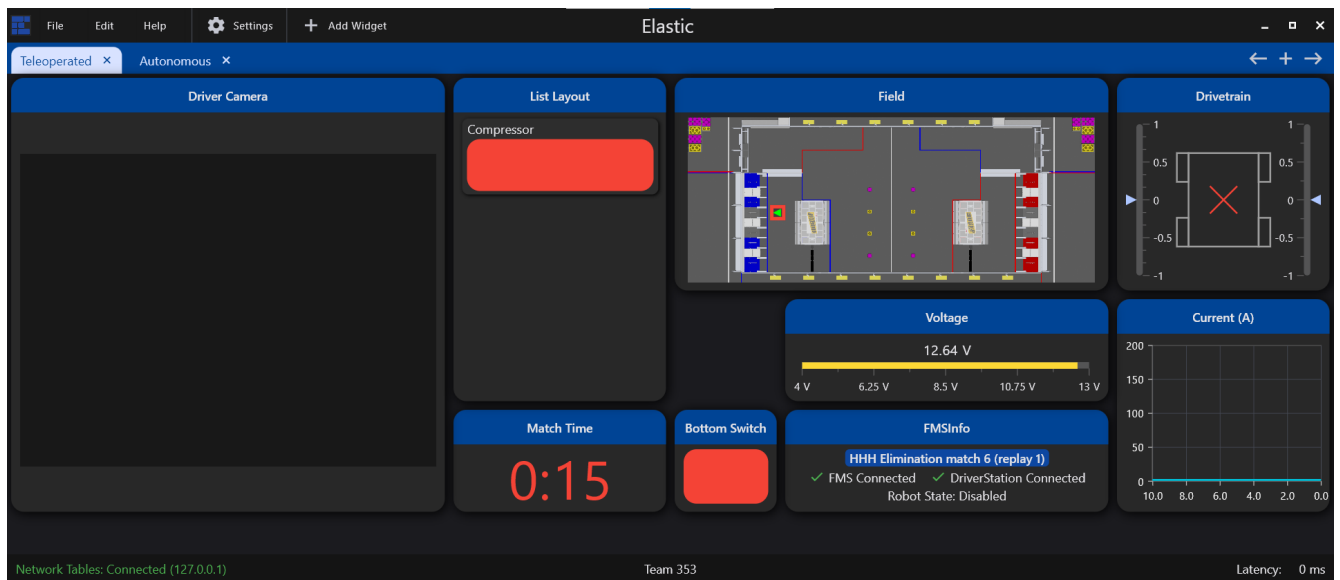
Adding Widgets to Layouts / 向布局中添加控件

To add a widget to a layout, simply drag an existing widget over the layout, and when your mouse cursor is hovering over the layout, it will be highlighted yellow. Letting go of your mouse cursor will place the widget into the layout.

要向布局中添加控件，只需将现有控件拖到布局上方。当鼠标光标悬停在布局上方时，布局会以黄色高亮显示。松开鼠标，即可将控件放入布局。



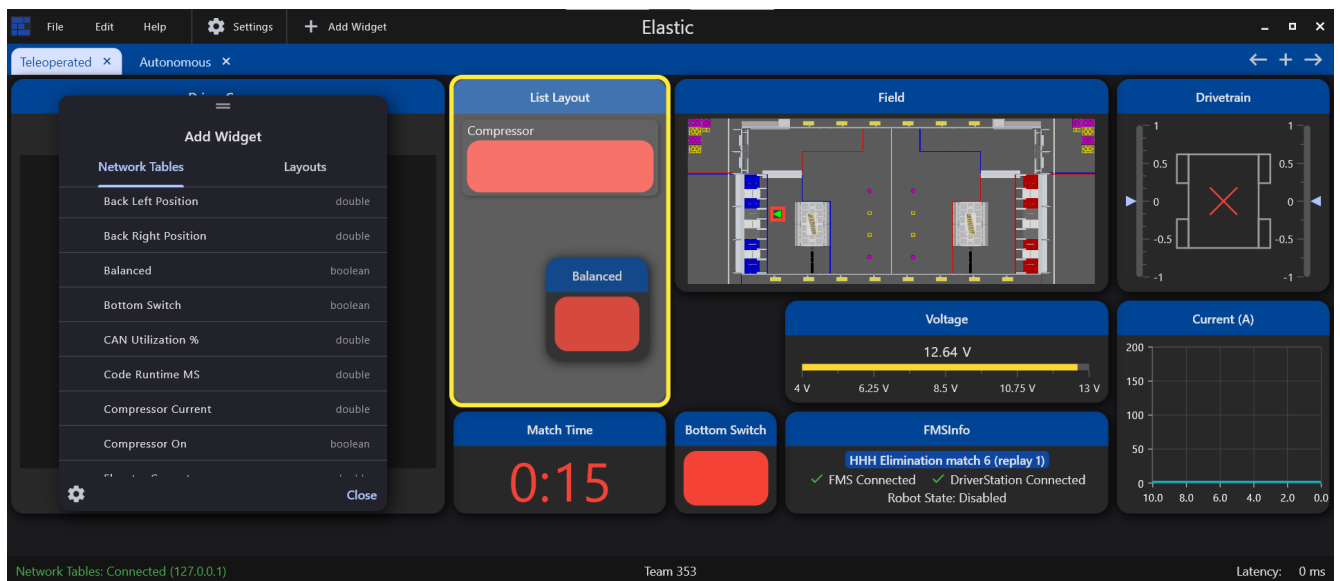
Dragging widget into layout / 将控件拖入布局



Widget in layout / 布局中的控件

Alternatively, you can drag a widget from the "Add Widget" dialog straight into the layout.

也可以从 "Add Widget" 对话框中直接将控件拖入布局。

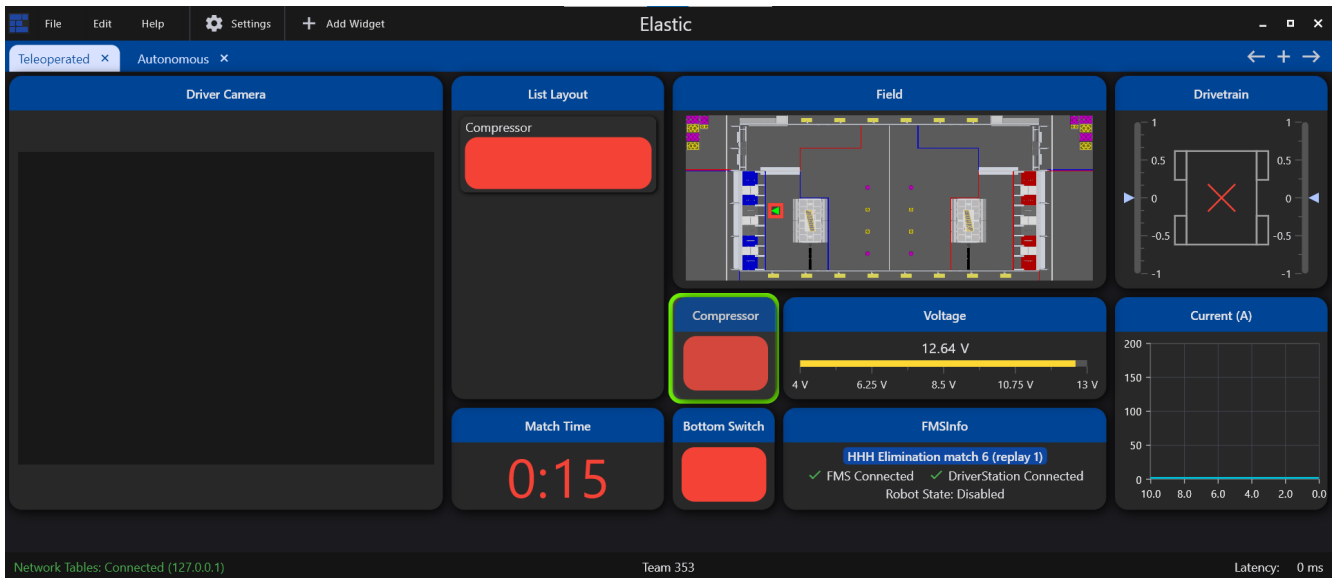


Dragging new widget into layout / 将新控件拖入布局

Moving Widgets Back onto Grid / 将控件移回网格

If you want to remove a widget from a layout and add it back to the grid, simply click and drag the widget out of the layout and back to where you want it on the grid.

如果要将控件从布局中移除并放回网格，只需点击并将控件从布局中拖出，再拖到网格中的目标位置。

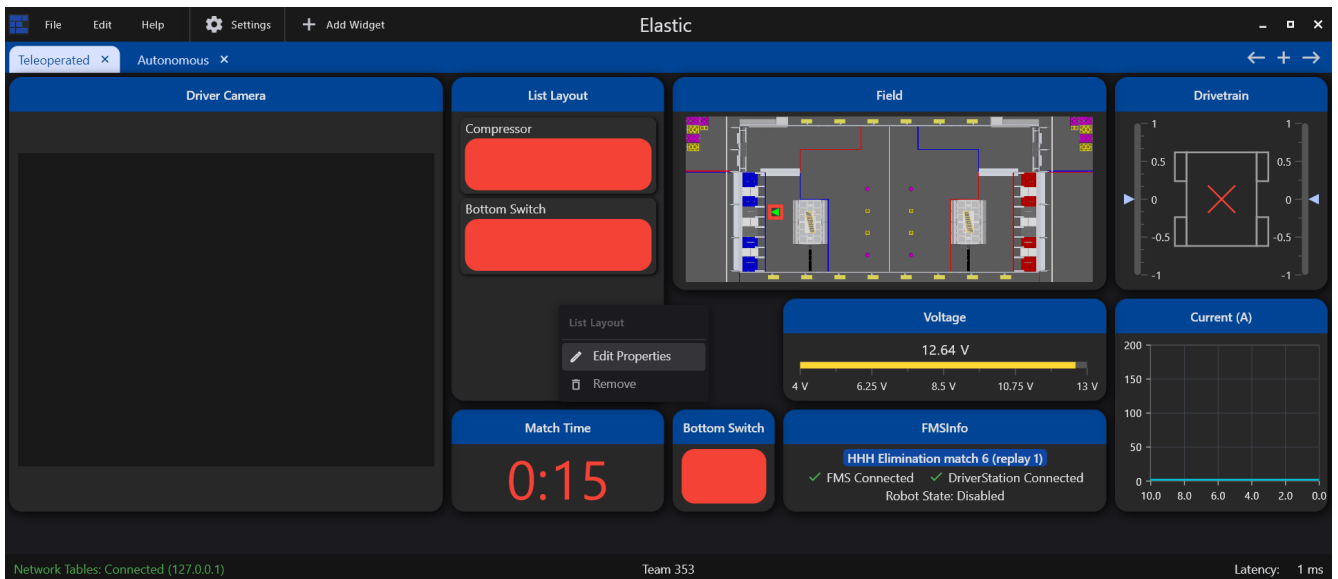


Dragging widget out of layout / 将控件拖出布局

Deleting Widgets From Layout / 从布局中删除控件

If you have a widget in a layout that you want to remove entirely, right click on the layout, then click "Edit Properties".

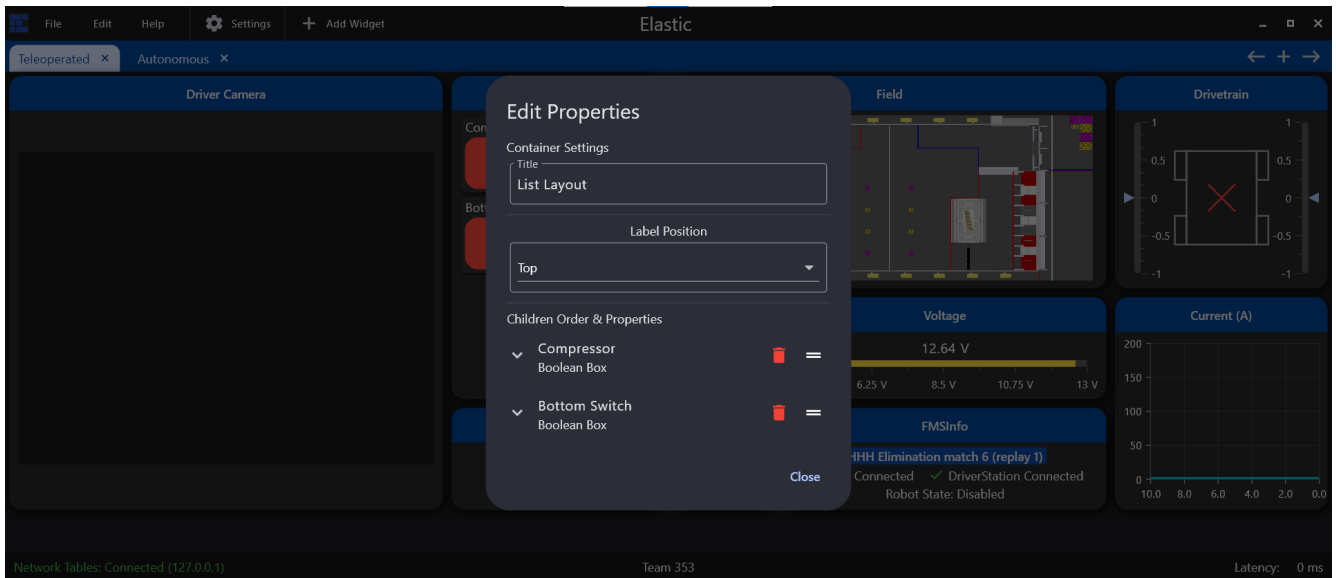
如果要彻底移除布局中的某个控件，请右键点击布局，然后点击 "Edit Properties".



Edit Properties Button / 编辑属性按钮

When you click edit properties, a menu will show up with settings to edit and reorder each element in the layout.

点击编辑属性后，会出现一个菜单，其中的设置可用于编辑布局中的各个元素并调整顺序。



Edit Properties Menu / 编辑属性菜单

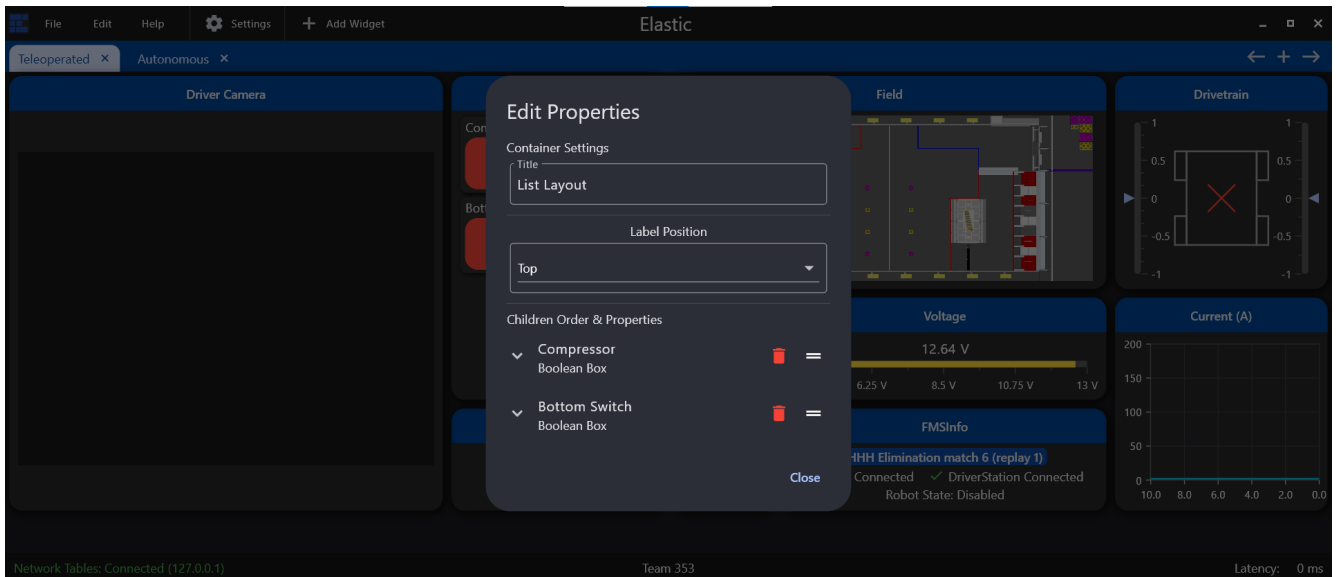
Clicking the trash can button next to the widget will remove that widget from the layout.

点击控件旁的垃圾桶按钮，会将该控件从布局中移除。

Editing the Order and Properties of Widgets in a Layout / 编辑布局中控件的顺序和属性

To change the order or properties of a widget in your layout, right click the layout, then click "Edit Properties", which will bring up a menu that looks like this.

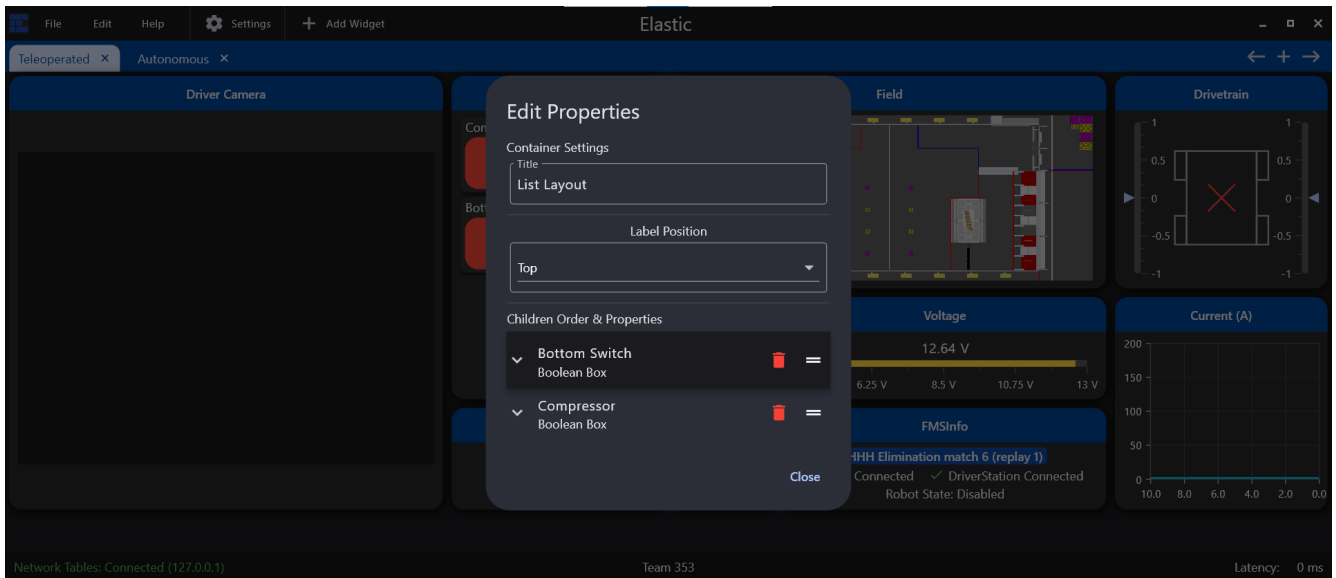
要修改布局中控件的顺序或属性，请右键点击布局，再点击 "Edit Properties"，随后会出现如下菜单。



Edit Properties Menu / 编辑属性菜单

To move a widget to a different spot in the layout, click and drag the handle on the right.

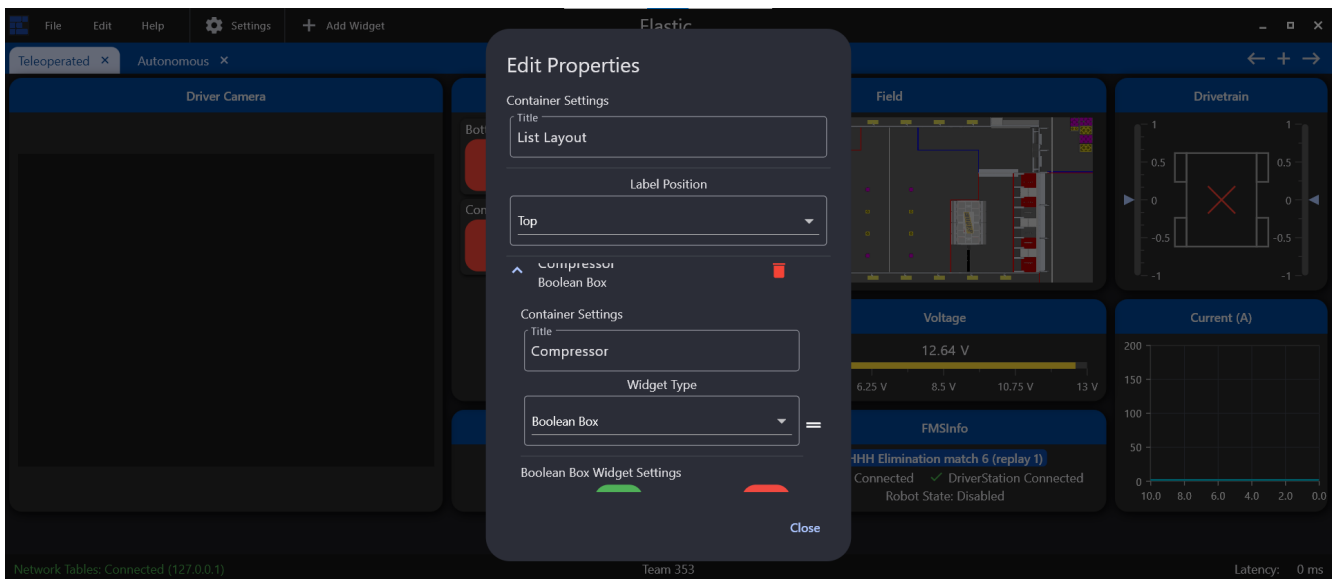
要将控件移动到布局中的其他位置，请点击并拖动右侧的拖动手柄。



Changing Order / 调整顺序

To change the properties of a widget that is in your layout, click the arrow on the left side of the tile for the widget. Doing so will reveal the editable properties of that specific widget.

要修改布局中控件的属性，请点击该控件条目左侧的箭头。这样会展开该控件可编辑的属性。

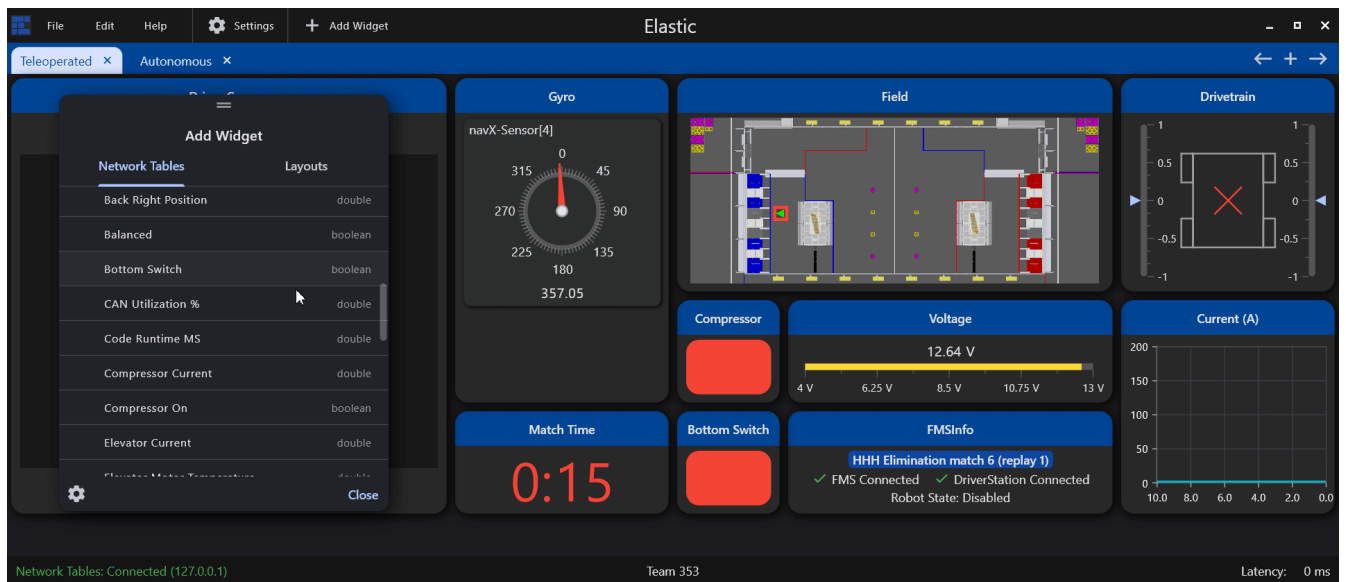


Widget Edit Properties / 控件编辑属性

Adding Gyro (Example) / 添加陀螺仪 (示例)

This gif is an example of how to add a layout. This adds a list layout and puts info about the gyro sensor in it, along with demonstrating how to change the settings.

此 GIF 展示了如何添加布局：添加一个列表布局，并在其中放入陀螺仪传感器的信息，同时演示如何修改设置。



Adding Gyro Example / 添加陀螺仪示例

Animated GIF: view original / 动画 GIF: 查看原动画

Widgets List & Properties Reference / 控件列表与属性参考

Information about Properties / 属性说明

Every widget has properties, with settings such as where to receive information, update frequency, and unique customization options. By default, every widget has 2 properties, `topic` (String), and `period` (double).

每个控件都有属性，包括数据来源、更新频率以及专有的定制选项。默认情况下，每个控件都有 2 个属性：`topic` (String) 和 `period` (double)。

The `topic` property specifies where to stream data from on NetworkTables, and `period` determines the update frequency of the widget.

`topic` 属性指定从 NetworkTables 中何处获取数据流，`period` 决定控件的更新频率。

Single-Topic Widgets / 单 topic 控件

These are widgets that only display data from one topic. They subscribe to the topic and display the raw data sent in the desired format.

这些控件只显示一个 topic 的数据。它们订阅该 topic，并按所需格式显示接收到的原始数据。

Text Display / 文本显示

Displays data in a text field. This text can be edited to update the data on Network Tables, it supports displaying and editing the following data types:

在文本框中显示数据。可以编辑文本来更新 NetworkTables 中的数据，支持显示和编辑以下数据类型：

- `boolean`
- `float`
- `double`
- `int`
- `float[]`
- `double[]`
- `int[]`
- `boolean[]`

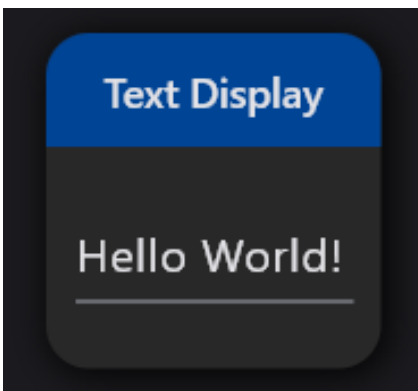


image / 图像

Properties / 属性

- `show_submit_button` - Whether or not to show a button to publish the entered data to network tables (defaults to false)

`show_submit_button` - 是否显示用于将输入数据发布到 NetworkTables 的按钮（默认为 false）

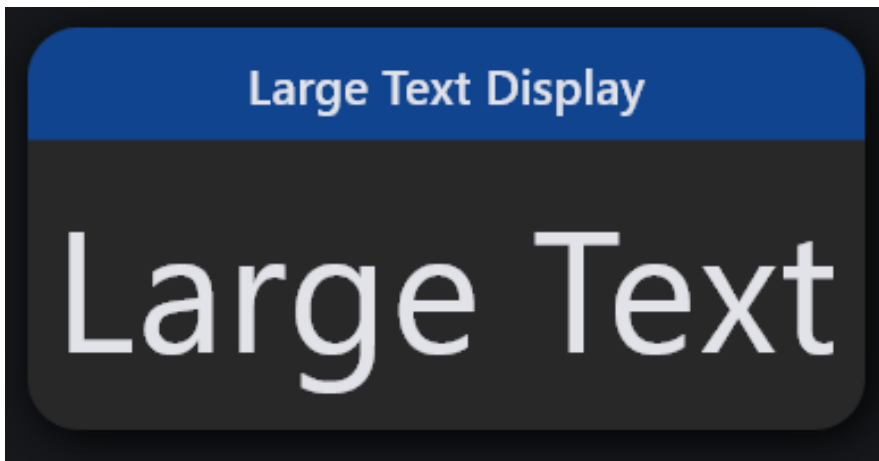
- `true` OR `false`

`true` 或 `false`

Large Text Display / 大号文本显示

Displays any String or number in large text that will fill the entire widget.

以填满整个控件的大号文本显示任意 String 或数值。



Number Slider / 数值滑块

Displays a number in the form of a slider. The slider can be dragged to edit the value on Network Tables. It can be customized with different ranges and intervals. This supports the `double` data type.

以滑块形式显示数值。可以拖动滑块来编辑 NetworkTables 中的值。可定制不同的范围和间隔。支持 `double` 数据类型。

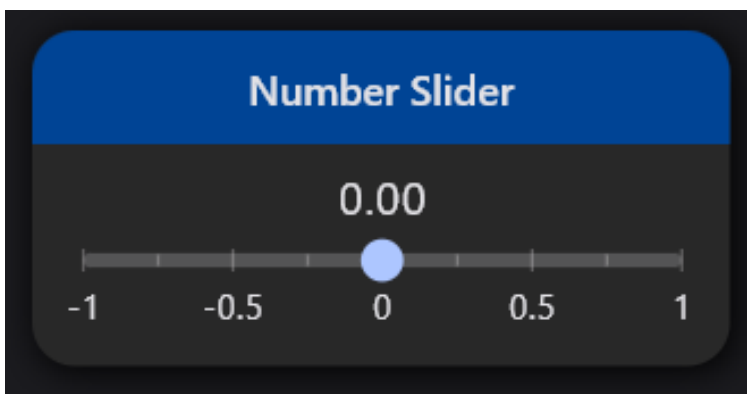


image / 图像

Properties / 属性

- `min_value` - The minimum value of the number slider (defaults to -1.0)

`min_value` - 数值滑块的最小值（默认为 -1.0）

- Decimal number

小数

- `max_value` - The maximum value of the number slider (defaults to 1.0)

`max_value` - 数值滑块的最大值 (默认为 1.0)

- Decimal number

小数

- `divisions` - The number of divisions between the minimum and maximum (defaults to 5)

`divisions` - 最小值和最大值之间的分段数量 (默认为 5)

- Positive whole number

正整数

- `publish_all` - Whether or not to publish the slider value when dragging (defaults to false)

`publish_all` - 是否在拖动时发布滑块的值 (默认为 false)

- true OR false

true 或 false

Number Bar / 数值条

Similar to the Number Slider, but does not support editing values.

与数值滑块类似，但不支持编辑数值。

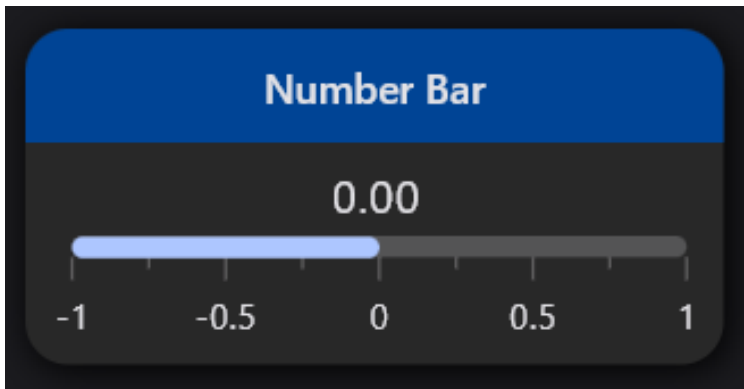


image / 图像

Properties / 属性

- `min_value` - The minimum value of the number bar (defaults to -1.0)

`min_value` - 数值条的最小值 (默认为 -1.0)

- Decimal number

小数

- `max_value` - The maximum value of the number bar (defaults to 1.0)

`max_value` - 数值条的最大值 (默认为 1.0)

- Decimal number

小数

- divisions - The number of divisions between the minimum and maximum (defaults to 5)

divisions - 最小值和最大值之间的分段数量 (默认为 5)

- Positive whole number

正整数

- inverted - Whether or not the bar should be inverted (defaults to false)

inverted - 是否反转数值条 (默认为 false)

- true OR false

true 或 false

- orientation - The bar's orientation (defaults to horizontal)

orientation - 数值条的方向 (默认为 horizontal)

- horizontal OR vertical (String)

horizontal 或 vertical (String)

Voltage View / 电压显示

Displays a yellow number bar with the V units next to it, which is particularly useful for viewing the robot's voltage.

显示黄色数值条，并在旁边显示单位 V，尤其适合查看机器人的电压。

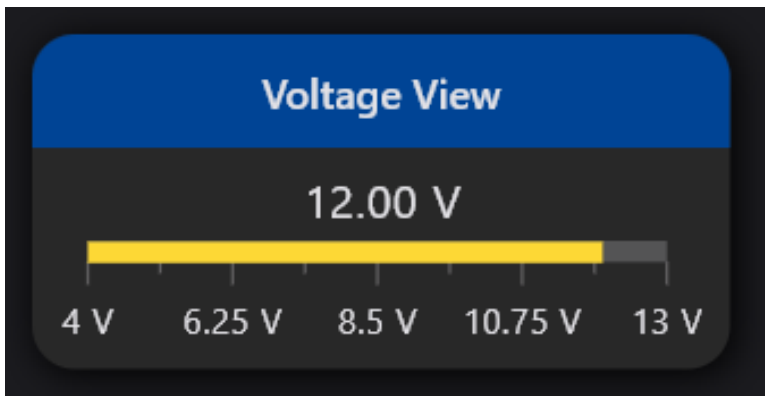


image / 图像

Properties / 属性

- min_value - The minimum value of the number bar (defaults to 4.0)

min_value - 数值条的最小值 (默认为 4.0)

- Decimal number

小数

- max_value - The maximum value of the number bar (defaults to 13.0)

max_value - 数值条的最大值 (默认为 13.0)

- Decimal number

小数

- divisions - The number of divisions between the minimum and maximum (defaults to 5)

divisions - 最小值和最大值之间的分段数量 (默认为 5)

- Positive whole number

正整数

- inverted - Whether or not the bar should be inverted (defaults to false)

inverted - 是否反转数值条 (默认为 false)

- true OR false

true 或 false

- orientation - The bar's orientation (defaults to horizontal)

orientation - 数值条的方向 (默认为 horizontal)

- horizontal OR vertical (String)

horizontal 或 vertical (String)

Radial Gauge / 圆形仪表

Displays a number in a style similar to a car's speedometer.

以类似汽车速度表的样式显示数值。

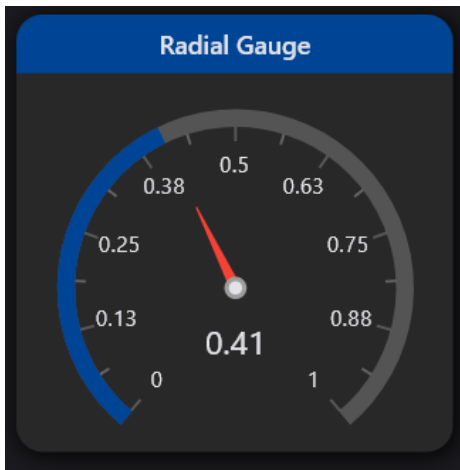


image / 图像

Properties / 属性

- start_angle - The angle (in degrees) to start at (defaults to -140.0)

start_angle - 起始角度 (单位为度, 默认为 -140.0)

- Decimal number

小数

- end_angle - The angle (in degrees) to end at (defaults to 140.0)

end_angle - 结束角度 (单位为度, 默认为 140.0)

- Decimal number

小数

- `min_value` - The lowest value to display (defaults to 0.0)
`min_value` - 显示的最小值 (默认为 0.0)
- Decimal number
小数
- `max_value` - The highest value to display (defaults to 100.0)
`max_value` - 显示的最大值 (默认为 100.0)
- Decimal number
小数
- `number_of_labels` - The number of labels to display on the gauge (defaults to 8)
`number_of_labels` - 仪表上显示的标签数量 (默认为 8)
- Positive whole number
正整数
- `wrap_value` - Whether or not to wrap the value between the min and max, particularly useful for displaying gyro values (defaults to false)
`wrap_value` - 是否将数值循环映射到最小值和最大值之间, 尤其适合显示陀螺仪数值 (默认为 false)
- true OR false
true 或 false
- `show_pointer` - Whether or not to show a needle pointing towards the value (defaults to true)
`show_pointer` - 是否显示指向当前数值的指针 (默认为 true)
- true OR false
true 或 false
- `show_ticks` - Whether or not to show tick marks (defaults to true)
`show_ticks` - 是否显示刻度线 (默认为 true)
- true OR false
true 或 false

Graph / 图表

Displays a Graph of how data has been changing over time. This can display a line in different colors and over different time intervals.

显示数据随时间变化的图表。可以使用不同颜色和不同时间区间显示曲线。

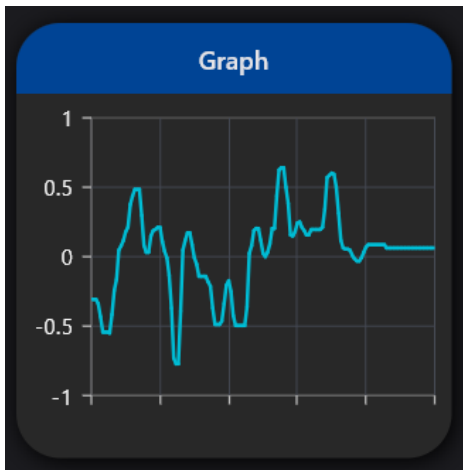


image / 图像

Properties / 属性

- `time_displayed` - The amount of time (in seconds) to display the latest data (defaults to 5.0)
`time_displayed` - 显示最新数据的时长 (单位为秒, 默认为 5.0)
- `Decimal number`
 小数
- `min_value` - The minimum Y value of the graph (defaults to nothing)
`min_value` - 图表的最小 Y 值 (默认不设置)
- `Decimal number`
 小数
- If no value is provided, the graph will dynamically change the minimum Y value depending on the data the graph is displaying
 如果没有提供此值, 图表会根据当前显示的数据动态调整最小 Y 值
- `max_value` - The maximum Y value of the graph (defaults to nothing)
`max_value` - 图表的最大 Y 值 (默认不设置)
- `Decimal number`
 小数
- If no value is provided, the graph will dynamically change the minimum Y value depending on the data the graph is displaying
 如果没有提供此值, 图表会根据当前显示的数据动态调整最小 Y 值
- `color` - The color of the line (defaults to light blue)
`color` - 曲线的颜色 (默认为浅蓝色)
- `32 bit int (format: 0xAARRGGBB)`
 32 位 int (格式: 0xAARRGGBB)
- `line_width` - The width of the line in pixels (defaults to 2.0)
`line_width` - 曲线宽度, 单位为像素 (默认为 2.0)

- Positive decimal number

正小数

32 Bit int information

32 位 int 说明

- The value starts with 0x, and then the ARGB code

该值以 0x 开头，后面是 ARGB 代码

- To obtain the ARGB code, use this [Color Converter](#)

要获取 ARGB 代码，请使用此[颜色转换器](#)

Match Time / 比赛时间

Displays a very large number that is color coded to how much time is left in a match. Any time above 1 minute will be in blue, below 1 minute will be green, below 30 seconds will be yellow, and 15 and below will be red. This can display time in raw seconds or MM:SS format

以很大的数字显示比赛剩余时间，并根据剩余时间设置颜色。大于 1 分钟时显示蓝色，小于 1 分钟时显示绿色，小于 30 秒时显示黄色，15 秒及以下时显示红色。可以以原始秒数或 MM:SS 格式显示时间

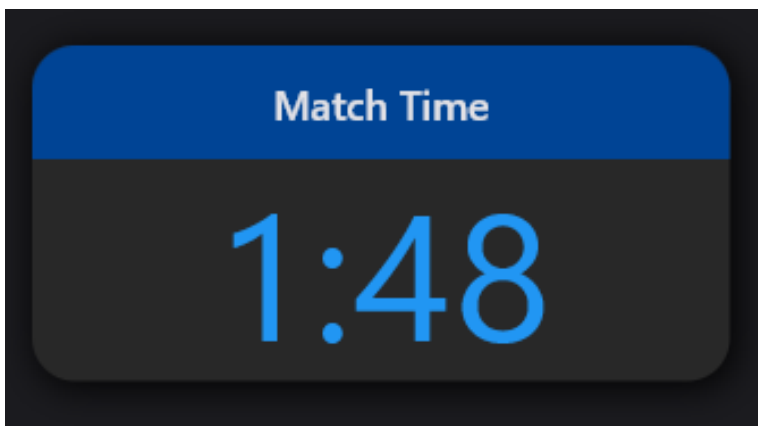


image / 图像

Properties / 属性

- `time_display_mode` - The format the time should be displayed in (defaults to "Minutes and Seconds")

`time_display_mode` - 时间显示格式（默认为 "Minutes and Seconds"）

- Minutes and Seconds OR Seconds Only

Minutes and Seconds 或 Seconds Only

- `red_start_time` - The time in seconds when the text will display in the color red (defaults to 15)

`red_start_time` - 文本开始显示红色的时间，单位为秒（默认为 15）

- Whole number

整数

- `yellow_start_time` - The time in seconds when the text will display in the color yellow (defaults to 30)

`yellow_start_time` - 文本开始显示黄色的时间，单位为秒（默认为 30）

- Whole number

整数

Boolean Box / 布尔值方框

The simplest of all the widgets. Displays a color when the value is true, and a different color when the value is false. These colors default to red and green, but can be changed as you wish.

最简单的控件。值为 true 时显示一种颜色，值为 false 时显示另一种颜色。这两种颜色默认为红色和绿色，但可以按需修改。



image / 图像

Properties / 属性

- true_color - the Color to display when the value is "true" (defaults to green)

true_color - 值为 "true" 时显示的颜色（默认为绿色）

- 32 bit int (format: 0xAARRGGBB)

32 位 int（格式：0xAARRGGBB）

- false_color - the color to display when the value is "false" (defaults to red)

false_color - 值为 "false" 时显示的颜色（默认为红色）

- 32 bit int (format: 0xAARRGGBB)

32 位 int（格式：0xAARRGGBB）

32 Bit int information

32 位 int 说明

- The value starts with 0x, and then the ARGB code

该值以 0x 开头，后面是 ARGB 代码

- To obtain the ARGB code, use this [Color Converter](#)

要获取 ARGB 代码，请使用此[颜色转换器](#)

Toggle Button / 切换按钮

Displays a boolean value in the form of a button. This button can be pressed to switch the boolean value from true to false and vice versa.

以按钮形式显示布尔值。按下按钮，可以将布尔值从 true 切换为 false，或从 false 切换为 true。

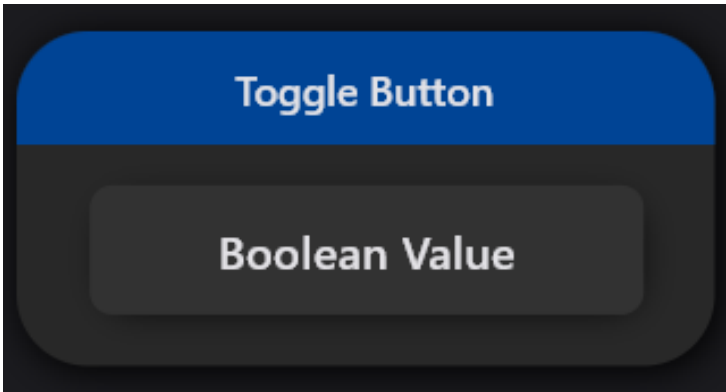


image / 图像

Toggle Switch / 切换开关

Displays a boolean value in the form of a switch. This switch can be toggled in order to invert the value.

以开关形式显示布尔值。切换此开关可反转该值。



image / 图像

Single Color View / 单色显示

This is a custom widget, for code examples on how to display this, see [Custom Widget Examples](#)

这是一个自定义控件。有关如何显示它的代码示例，请参阅[自定义控件示例](#)

This displays any color in a box. This can be very useful for seeing what a color sensor is detecting, or a color seen by a vision tracking camera.

在方框中显示任意颜色。这对于查看颜色传感器检测到的颜色，或视觉跟踪摄像头看到的颜色非常有用。



image / 图像

This widget will show as an option on any widget displaying data that is a String. In order for this to display a color, the string must be formatted as "#RRGGBB". It is recommended to use the `toHexString()` method in WPILib's Color class ([Java](#)) in order to obtain this hex code.

对于显示 String 数据的控件，此控件会作为一个可选项出现。要显示颜色，字符串必须采用 "#RRGGBB" 格式。建议使用 WPILib 的 Color 类中的 `toHexString()` 方法 ([Java](#)) 来获取此十六进制代码。

Multi Color View / 多色显示

Displays a gradient of multiple colors. Will display on a topic with the `string[]` type if every string is a color hex code.

显示由多种颜色组成的渐变。如果 `string[]` 类型的 topic 中每个字符串都是颜色十六进制代码，则可在该 topic 上显示。

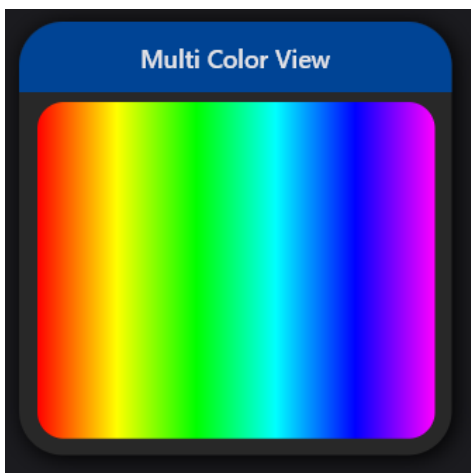


image / 图像

This widget will show as an option on any widget displaying data that is a String array. In order for this to display a color, the elements in the array must be formatted as "#RRGGBB". It is recommended to use the `toHexString()` method in WPILib's Color class ([Java](#)) in order to obtain these hex codes.

对于显示 String 数组数据的控件，此控件会作为一个可选项出现。要显示颜色，数组元素必须采用 "#RRGGBB" 格式。建议使用 WPILib 的 Color 类中的 `toHexString()` 方法 ([Java](#)) 来获取这些十六进制代码。

Multi-Topic Widgets / 多 topic 控件

These widgets display data from multiple topics. Most of these are built from Sendables in robot code, which is the recommended way of having these added to Network Tables.

这些控件显示来自多个 topic 的数据。大部分控件基于机器人代码中的 Sendable 构建，这也是将这些数据添加到 NetworkTables 的推荐方式。

3-Axis Accelerometer / 3 轴加速度计

Built from the WPILib BuiltInAccelerometer (Java), displays the X, Y, and Z acceleration of an accelerometer.

基于 WPILib BuiltInAccelerometer (Java) 构建，显示加速度计的 X、Y 和 Z 轴加速度。

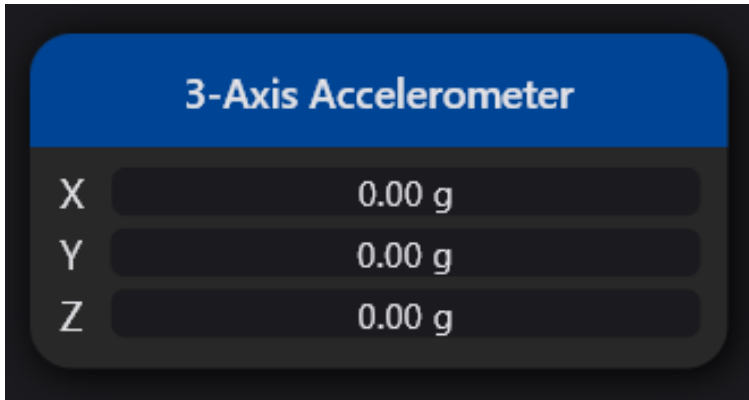


image / 图像

Accelerometer / 加速度计

Displays data from any 1-axis accelerometer.

显示任意 1 轴加速度计的数据。



image / 图像

Camera Stream / 摄像头视频流

Shows a live camera view from Camera Server. It is recommended to use a very low stream resolution as high resolution video can heavily affect performance.

显示来自 Camera Server 的实时摄像头画面。建议使用很低的视频流分辨率，因为高分辨率视频会严重影响性能。

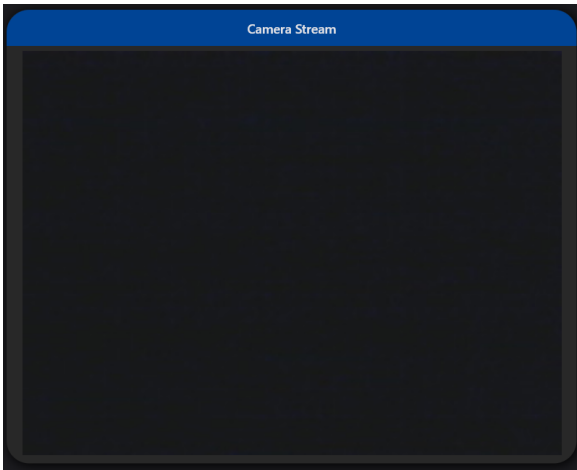


image / 图像

Properties / 属性

- `compression` - The quality of the image between 0 and 100, with 0 being lowest quality, and 100 being the highest quality, defaults to null

`compression` - 图像质量，范围为 0 到 100，其中 0 表示最低质量，100 表示最高质量，默认为 null

- Whole number

整数

- `fps` - How many frames per second should be sent from the server, defaults to null

`fps` - 服务器每秒应发送的帧数，默认为 null

- Whole number

整数

- `resolution` - The resolution to apply to the stream, defaults to null

`resolution` - 应用于视频流的分辨率，默认为 null

- Integer array, in the format `[width, height]`

整数数组，格式为 `[width, height]`

ComboBox Chooser / 下拉框选择器

Built from WPILib's Sendable Chooser ([Java](#)), provides a dropdown list of all of the options, and the ability to search for an option. If the name of it is too long, hovering over the widget will reveal a tooltip showing the full name of the selected option.

基于 WPILib 的 Sendable Chooser ([Java](#)) 构建，提供包含所有选项的下拉列表，并支持搜索选项。如果名称过长，将鼠标悬停在控件上，会出现显示所选选项完整名称的工具提示。

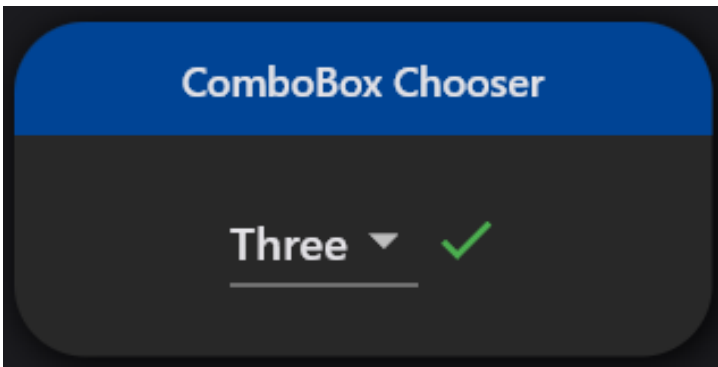


image / 图像

Properties / 属性

- `sort_options` - Whether or not to sort the options alphabetically, defaults to `false`
`sort_options` - 是否按字母顺序排列选项，默认为 `false`
- `trueOr false`
`true` 或 `false`

Split Button Chooser / 分组按钮选择器

Also built from WPILib's Sendable Chooser ([Java](#)), provides a list of all of the options as a row of buttons which can be scrolled left and right.

同样基于 WPILib 的 Sendable Chooser ([Java](#)) 构建，以一行按钮列出所有选项，并支持左右滚动。

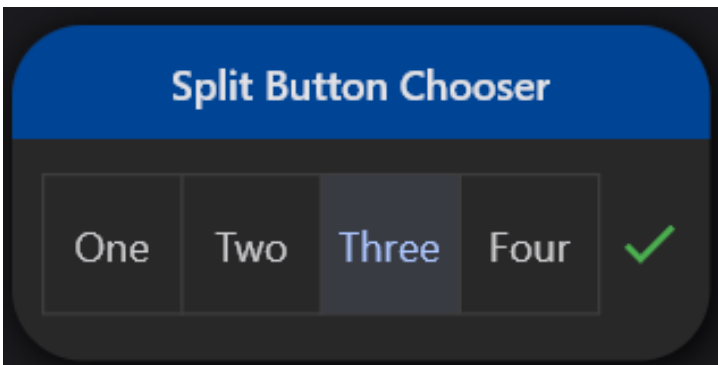


image / 图像

Command Scheduler / Command 调度器

Displays a list of all of the commands scheduled, along with their IDs and a button to unschedule them. This is created using WPILib's [CommandScheduler \(Java\)](#)

显示所有已调度 Command 的列表、它们的 ID，以及取消调度的按钮。此控件使用 WPILib 的 [CommandScheduler \(Java\)](#) 创建

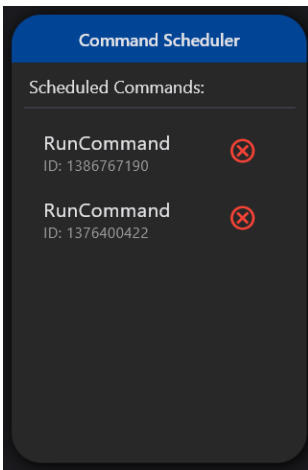


image / 图像

Command

Displays a button to run a command.

显示用于运行 Command 的按钮。

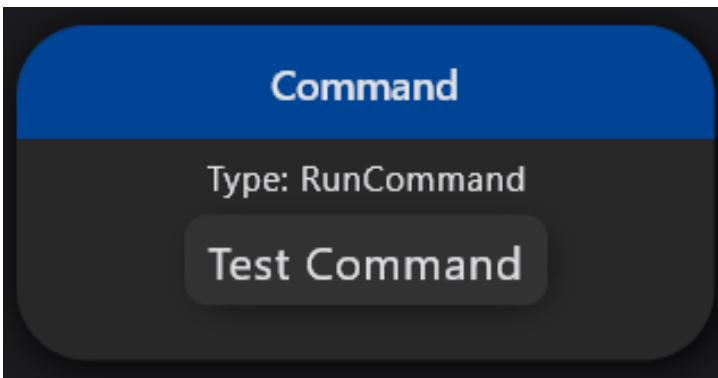


image / 图像

Properties / 属性

- `show_type` - Whether or not to show text that says the type of command (defaults to true)
`show_type` - 是否显示说明 Command 类型的文字（默认为 true）
- `true` OR `false`
`true` 或 `false`
- `maximize_button_space` - Whether or not to have the button fill the entire widget
`maximize_button_space` - 是否让按钮填满整个控件
- `true` OR `false`
`true` 或 `false`

Subsystem / subsystem

Displays information about a robot subsystem.

显示机器人 subsystem 的信息。

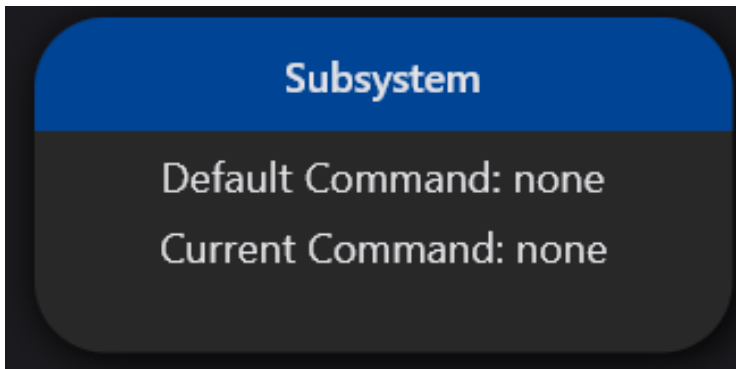


image / 图像

DifferentialDrive

Built from the WPILib DifferentialDrive ([Java](#)), it displays the wheel speeds of a differential drivetrain along with a curved arrow showing its direction. The sliders on the left and right can be dragged to set the speeds of each wheel (only works when Live Window is enabled).

基于 WPILib DifferentialDrive ([Java](#)) 构建，显示差速底盘的轮速，以及表示其运动方向的弯曲箭头。可以拖动左右两侧的滑块来设置各车轮速度（仅在 Live Window 启用时有效）。



image / 图像

SwerveDrive

This is a custom widget, for code examples on how to display this, see [Custom Widget Examples](#)

这是一个自定义控件。有关如何显示它的代码示例，请参阅[自定义控件示例](#)

Displays the wheel rotation and velocity of a swerve drivetrain, along with the robot's rotation. There is also a similar widget that can display data from any robot program using the YAGSL library more info in the [Custom Widget Examples](#) page.

显示 swerve 底盘各车轮的旋转角度和速度，以及机器人的旋转角度。另有一个类似控件，可显示任意使用 YAGSL 库的机器人程序的数据；更多信息见[自定义控件示例](#)页面。

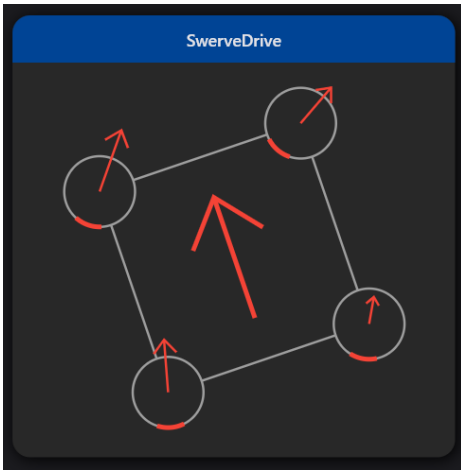


image / 图像

Properties / 属性

- `show_robot_rotation` - whether or not to rotate the robot in the widget (defaults to true)
`show_robot_rotation` - 是否旋转控件中的机器人 (默认为 true)
 - true OR false
true 或 false
- `rotation_unit` - The unit to use for the rotation of the modules and robot (defaults to "Radians")
`rotation_unit` - 模块和机器人旋转角度使用的单位 (默认为 "Radians")
 - Must be one of the following:
必须为以下值之一:
 - Radians
 - Degrees
 - Rotations
 - This option is not available on the YAGSL widget, as the units are handled automatically via Network Tables
 YAGSL 控件中没有此选项, 因为单位通过 NetworkTables 自动处理

Encoder / 编码器

Built from the WPILib Encoder ([Java](#)), displays the position and speed of an encoder.

基于 WPILib Encoder ([Java](#)) 构建, 显示编码器的位置和速度。

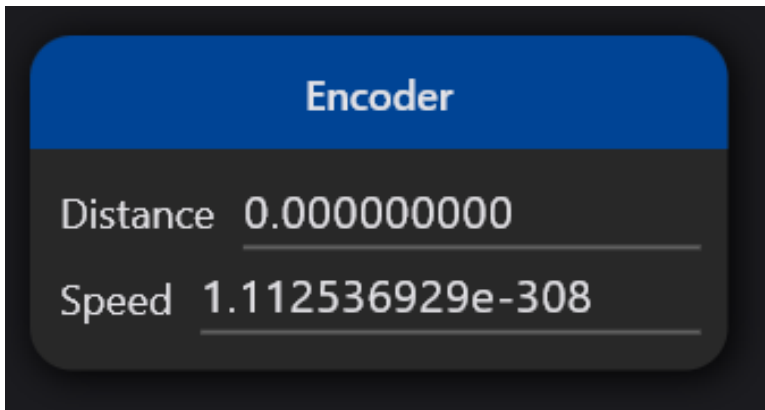


image / 图像

Field / 场地

Built from the WPILib Field2d ([Java](#)), displays a field with the robot's position and any other objects to scale. If a field object contains more than 8 elements, it will be displayed as a trajectory, which draws a line representing a path.

基于 WPILib Field2d ([Java](#)) 构建，按比例显示场地上的机器人位置及其他对象。如果一个场地对象包含超过 8 个元素，则会以轨迹形式显示，用一条线表示路径。

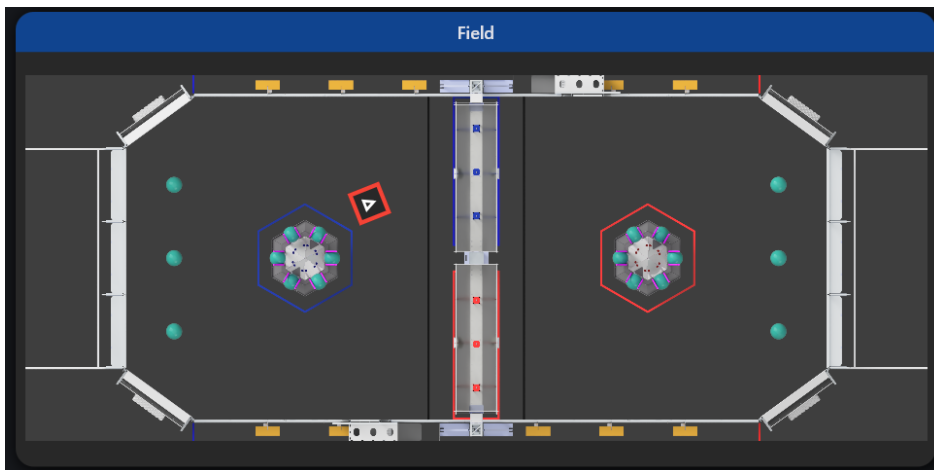


image / 图像

Properties / 属性

- `field_game` - The game's field to show (defaults to "Rebuilt")

`field_game` - 要显示的比赛场地（默认为 "Rebuilt"）

- Must be one of the following (case sensitive)

必须为以下值之一（区分大小写）

- Rebuilt
- Reefscape
- Crescendo
- Charged Up
- Rapid React

- Infinite Recharge
- Destination: Deep Space
- Power Up
- robot_width - The width of the robot in meters (defaults to 0.85)
robot_width - 机器人的宽度, 单位为米 (默认为 0.85)
- Decimal number
小数
- robot_length - The length of the robot in meters (defaults to 0.85)
robot_length - 机器人的长度, 单位为米 (默认为 0.85)
- Decimal number
小数
- show_other_objects - Whether or not to show other field objects that are not trajectories (defaults to true)
show_other_objects - 是否显示轨迹以外的其他场地对象 (默认为 true)
- true OR false
true 或 false
- show_trajectories - Whether or not to show trajectories (defaults to true)
show_trajectories - 是否显示轨迹 (默认为 true)
- true OR false
true 或 false
- field_rotation- How many degrees to rotate the field, clockwise positive (defaults to 0.0)
field_rotation - 场地旋转的角度, 单位为度, 顺时针为正 (默认为 0.0)
- Decimal number
小数
- robot_color- The color of the robot bumpers (defaults to red)
robot_color - 机器人 bumper 的颜色 (默认为红色)
- 32 bit int (format: 0xAARRGGBB)
32 位 int (格式: 0xAARRGGBB)
- trajectory_color - The color of the trajectory (defaults to white)
trajectory_color - 轨迹的颜色 (默认为白色)
- 32 bit int (format: 0xAARRGGBB)
32 位 int (格式: 0xAARRGGBB)
- show_robot_outside_widget - Whether or not to display the robot outside the widget (defaults to true)
show_robot_outside_widget - 是否在控件范围外显示机器人 (默认为 true)
- true OR false
true 或 false

32 Bit int Information

32 位 int 说明

- The value starts with 0x, and then the ARGB code
该值以 0x 开头，后面是 ARGB 代码
- To obtain the ARGB code, use this [Color Converter](#)
要获取 ARGB 代码，请使用此[颜色转换器](#)

FMSInfo

Automatically published to Network Tables by WPILib, displays information about the match, whether or not the robot is enabled, and other info such as FMS connection status and which alliance you are on.

由 WPILib 自动发布到 NetworkTables，显示比赛信息、机器人是否启用，以及 FMS 连接状态和所属联盟等其他信息。

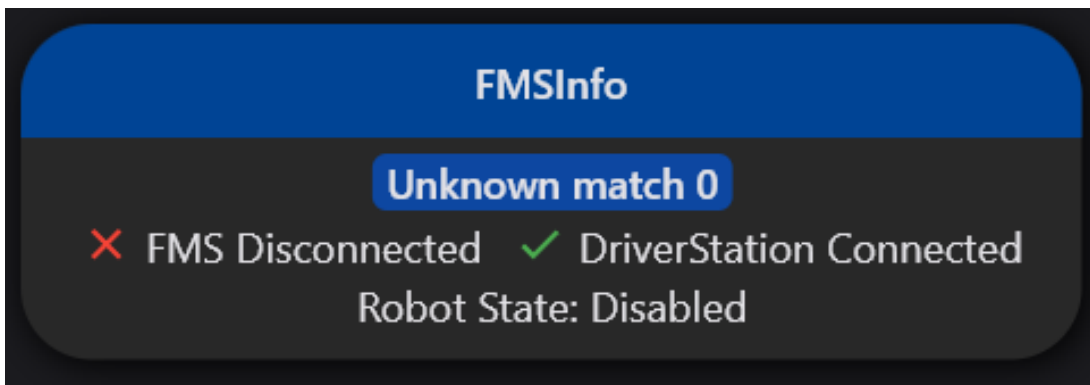


image / 图像

Gyro / 陀螺仪

Displays the robot's rotation in the form of a compass-like circular dial.

以类似指南针的圆形表盘显示机器人的旋转角度。



image / 图像

Properties / 属性

- counter_clockwise_positive - Whether or not the value of the Gyro is CCW+ (Defaults to false)

counter_clockwise_positive - 陀螺仪的数值是否以逆时针为正（默认为 false）

- true OR false

true 或 false

RobotPreferences

Displays anything under the /Preferences table when using the Preferences API. Anything submitted in the text boxes will be saved persistent in the Preferences table.

使用 Preferences API 时，显示 /Preferences 表下的全部内容。文本框中提交的所有内容都会以持久化方式保存在 Preferences 表中。

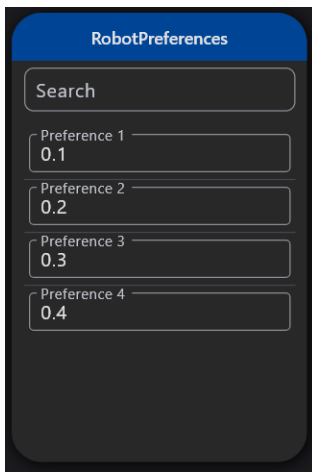
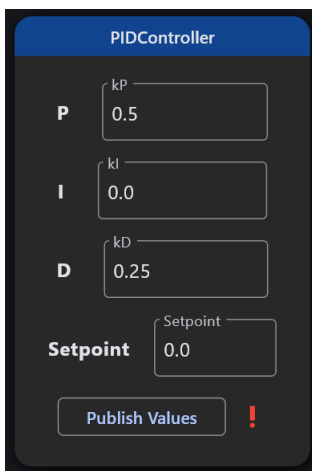


image / 图像

PIDController

Built from the WPILib PIDController ([Java](#)), allows editing of the gains and setpoint of a PID controller.

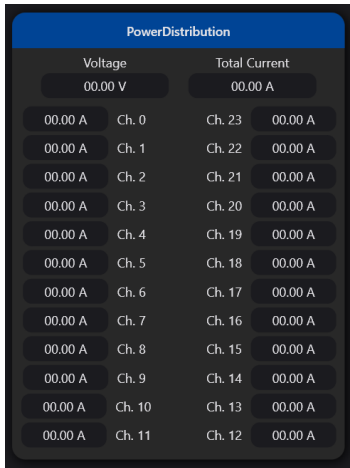
基于 WPILib PIDController ([Java](#)) 构建，允许编辑 PID 控制器的增益和 setpoint。



PowerDistribution

Built from the WPILib PowerDistribution ([Java](#)), displays the voltage, total current, and current drawn by all channels in a Power Distribution Panel/Hub.

基于 WPILib PowerDistribution ([Java](#)) 构建，显示 Power Distribution Panel/Hub 的电压、总电流以及所有通道消耗的电流。



PowerDistribution			
Voltage		Total Current	
00.00 V		00.00 A	
00.00 A	Ch. 0	Ch. 23	00.00 A
00.00 A	Ch. 1	Ch. 22	00.00 A
00.00 A	Ch. 2	Ch. 21	00.00 A
00.00 A	Ch. 3	Ch. 20	00.00 A
00.00 A	Ch. 4	Ch. 19	00.00 A
00.00 A	Ch. 5	Ch. 18	00.00 A
00.00 A	Ch. 6	Ch. 17	00.00 A
00.00 A	Ch. 7	Ch. 16	00.00 A
00.00 A	Ch. 8	Ch. 15	00.00 A
00.00 A	Ch. 9	Ch. 14	00.00 A
00.00 A	Ch. 10	Ch. 13	00.00 A
00.00 A	Ch. 11	Ch. 12	00.00 A

image / 图像

Relay / 继电器

Built from the WPILib Relay ([Java](#)), displays the current state of the relay, and allows it to be changed by selecting the different options from the list.

基于 WPILib Relay ([Java](#)) 构建，显示继电器的当前状态，并允许从列表中选择不同选项来修改状态。

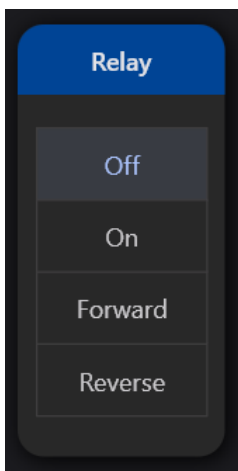


image / 图像

Ultrasonic / 超声波

Built from the WPILib Ultrasonic ([Java](#)), displays the distance reading from an Ultrasonic widget in inches.

基于 WPILib Ultrasonic ([Java](#)) 构建，以英寸为单位显示 Ultrasonic 控件的距离读数。

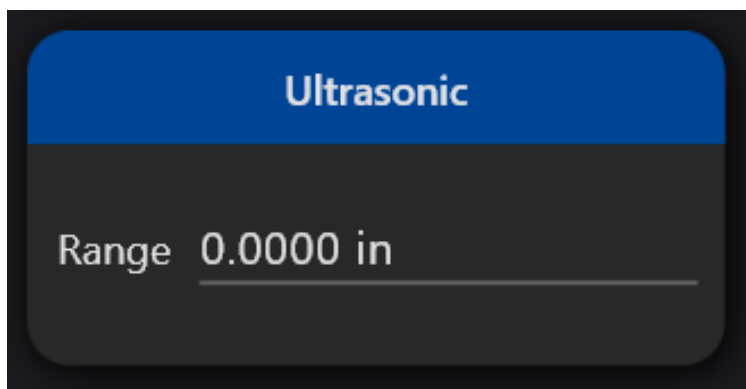


image / 图像

Alerts / 警报

Displays data from WPILib [Network Alerts](#).

显示来自 WPILib [Network Alerts](#) 的数据。

NetworkAlerts is being added to WPILib in 2025, robot code using a WPILib version from pre-2025 beta will not have access to the NetworkAlerts API.

NetworkAlerts 将在 2025 年加入 WPILib，使用 2025 beta 之前版本的 WPILib 的机器人代码无法使用 NetworkAlerts API。

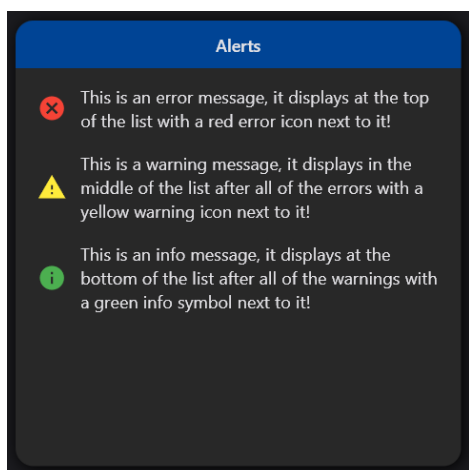


image / 图像

ElasticLib

Installing ElasticLib / 安装 ElasticLib

For Java projects, copy this file into your robot project:

<https://github.com/Gold872/elastic-dashboard/blob/main/elasticlib/Elastic.java>

对于 Java 项目，将此文件复制到机器人项目中：<https://github.com/Gold872/elastic-dashboard/blob/main/elasticlib/Elastic.java>

It is recommended to put this in a folder called `util`, however any location within a robot project works.

Depending on where the file is located, you may need to change the top line of the file.

建议将其放在名为 `util` 的文件夹中，但机器人项目内的任何位置都可以。根据文件所在位置，可能需要修改文件的第一行。

Robot Notifications / 机器人通知

Elastic supports sending notifications to the dashboard via robot code. This could be helpful in situations where you want to grab the attention of the user when something goes wrong or if there's an important warning to display.

Elastic 支持通过机器人代码向 Dashboard 发送通知。当出现问题需要引起用户注意，或需要显示重要警告时，此功能会很有帮助。

Sending notifications via robot code requires the use of ElasticLib. Java is supported.

通过机器人代码发送通知需要使用 ElasticLib。支持 Java。

Creating a Notification / 创建通知

Notification data is stored in an object called `Notification`. Currently, this has the following properties:

通知数据存储在名为 `Notification` 的对象中。目前，它具有以下属性：

- `level` for the type of notification
`level`: 通知类型
- `title` for the notification title
`title`: 通知标题
- `description` for the notification text
`description`: 通知文本
- `width` for the notification width (optional)
`width`: 通知宽度 (可选)
- `height` for the notification height (optional)
`height`: 通知高度 (可选)
- `displayTimeMillis` for the time to show the notification in milliseconds (optional)
`displayTimeMillis`: 通知显示时长，单位为毫秒 (可选)

There are 3 notification levels:

共有 3 种通知级别：

1. Error

错误

2. Warning

警告

3. Info

信息

An example of a Notification for an error notification would be

用于错误通知的 Notification 示例如下

Java

```
Elastic.Notification notification = new Elastic.Notification(Elastic.NotificationLevel.ERROR, "Error Notification", "This is an example error notification.");
```

Sending a notification / 发送通知

In order to send a notification, there is a method called `sendNotification` in the `Elastic` class to send a Notification.

`Elastic` 类中有一个名为 `sendNotification` 的方法，用于发送 Notification。

To send the error notification that was declared above, you would call

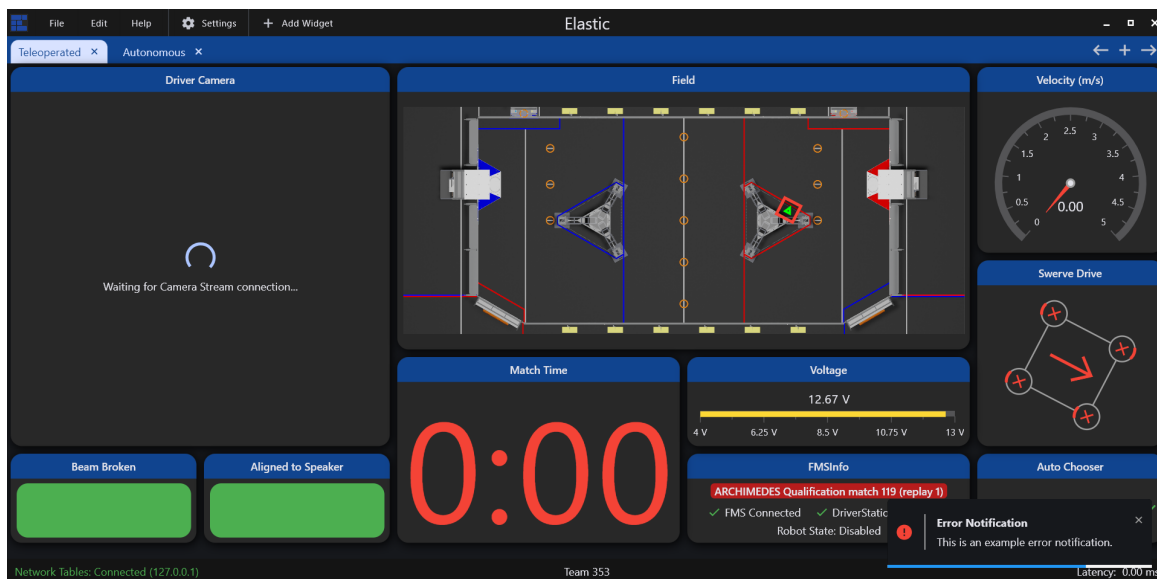
要发送上面声明的错误通知，可以调用

Java

```
Elastic.sendNotification(notification);
```

When this is called, a popup will appear on the dashboard that looks like this

调用此方法时，Dashboard 上会出现如下弹窗



Error Notification / 错误通知

Customizing a Notification / 定制通知

Notifications and their settings can be customized after being created, allowing them to be reused.

通知及其设置可以在创建后进行定制，从而复用通知。

For example, the error notification created above can be customized into a warning notification.

例如，可以将上面创建的错误通知定制为警告通知。

Java

```
/* code that created the notification */
notification.setLevel(NotificationLevel.WARNING);
notification.setTitle("Warning Notification");
notification.setDescription("This is an example warning notification");
```

Customizing with Method Chaining (Java Only) / 使用方法链定制通知（仅 Java）

For Java, method chaining is the simpler and recommended way to customize notifications. This allows for customizing multiple properties with just one line of code, as well as reusing only one notification object, which is better for the Java Garbage Collector performance (see [WPILib docs](#) for more details)

对于 Java，方法链是更简单且推荐的通知定制方式。它允许用一行代码定制多个属性，并且只复用一个通知对象，有利于 Java 垃圾回收器的性能（详情参阅 [WPILib 文档](#)）

For example, here's how a notification can be entirely customized and sent with just one line of code.

例如，下面演示了如何仅用一行代码完成通知的全部定制并发送。

```
Elastic.Notification notification = new Elastic.Notification();

/* ... */

Elastic.sendNotification(notification
    .withLevel(Elastic.NotificationLevel.INFO)
    .withTitle("Some Information")
    .withDescription("Your robot is doing fantastic!")
    .withDisplaySeconds(5.0)
);
```

Tab Switching / 切换选项卡

ElasticLib has functions to request the dashboard to switch the tab remotely.

ElasticLib 提供了请求 Dashboard 远程切换选项卡的函数。

To switch to a tab, use the `selectTab` method

要切换到某个选项卡，请使用 `selectTab` 方法

Java

```
Elastic.selectTab("Example Tab");
```

Additionally, you can also select a tab at a specific index (with index 0 being the first tab). To do this, pass an integer into the `selectTab` method.

此外，还可以选择指定索引的选项卡（索引 0 表示第一个选项卡）。要这样做，请将一个整数传入 `selectTab` 方法。

Java

```
// Select the tab at index 3 (the 4th tab)
Elastic.selectTab(3);
```

Remote Layout Downloading / 远程下载布局

For situations where you want to load a dashboard layout from your robot, Elastic supports downloading a layout remotely and merging it with your current layout.

如果希望从机器人加载 Dashboard 布局，Elastic 支持远程下载布局，并将其与当前布局合并。

On-Robot Configuration / 机器人端配置

In your robot code, add the following line in `robotInit()` or another initialization function

在机器人代码的 `robotInit()` 或其他初始化函数中，添加以下这行代码

Java

```
WebServer.start(5800, Filesystem.getDeployDirectory().getPath());
```

Next, you will have to deploy the Elastic configuration file to your robot:

接下来，需要将 Elastic 配置文件部署到机器人上：

1. Export the layout from Elastic

从 Elastic 导出布局

2. Move the exported json file into the deploy folder of your robot project

将导出的 json 文件移动到机器人项目的 deploy 文件夹中

3. Deploy your robot code

部署机器人代码

NOTE

说明

The exported file must end in ".json" and should be placed in the root directory of your deploy folder, the deploy folder structure should look like this:

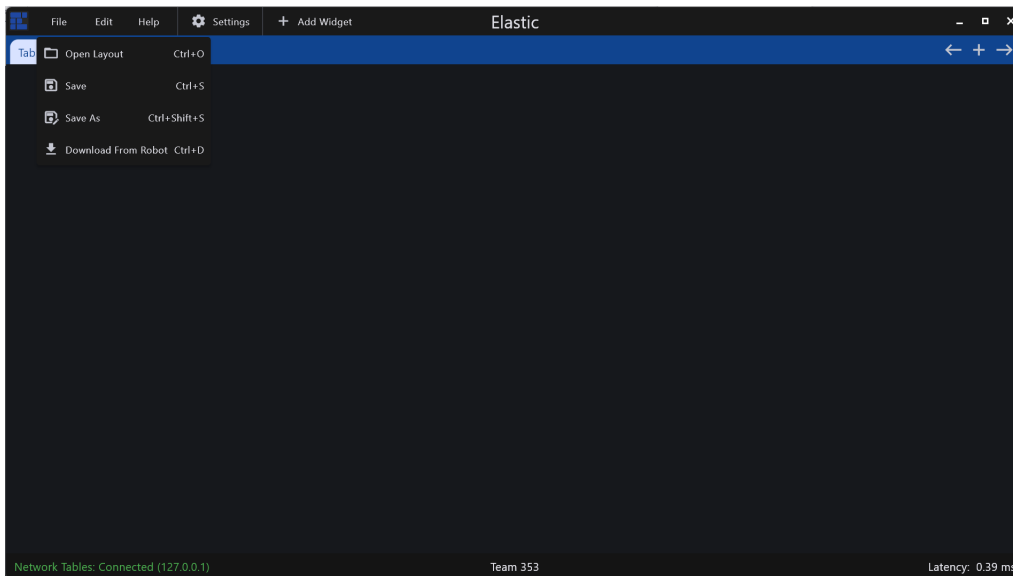
导出的文件必须以 ".json" 结尾，并应放在 deploy 文件夹的根目录中。deploy 文件夹结构应如下所示：

```
/deploy
elastic-layout.json
...
```

Downloading Your Layout / 下载布局

To download the layout to your dashboard, enter the File menu, and click Load Layout From Robot

要将布局下载到 Dashboard，请进入 File 菜单，并点击 Load Layout From Robot

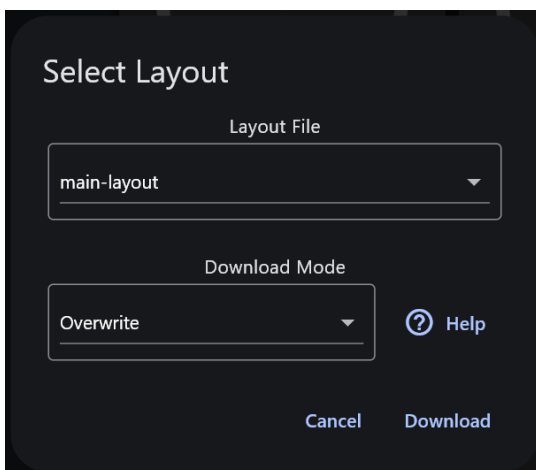


Alternatively, you can use the shortcut `Ctrl + D`

也可以使用快捷键 `Ctrl + D`

A dialog will appear with 2 dropdown menus, one to select which layout to download, and another to select how you want to download the data.

随后会出现一个包含 2 个下拉菜单的对话框，一个用于选择要下载的布局，另一个用于选择如何下载数据。



NOTE

说明

The layout file dropdown will display the name of any file in the root of your robot project deploy directory that ends with `.json`, if you aren't seeing your layout, ensure that your file is in the root of your deploy directory and the file ends with `.json`

布局文件下拉菜单会显示机器人项目 deploy 根目录中所有以 `.json` 结尾的文件名称。如果看不到布局，请确认文件位于 deploy 根目录中，且文件以 `.json` 结尾

Download Modes / 下载模式

Overwrite

覆盖

Keeps existing tabs that are not defined in the remote layout. However, any tabs that are defined in the remote layout will be overwritten locally.

保留远程布局中未定义的现有选项卡。不过，远程布局中定义的所有选项卡都会在本地被覆盖。

For example, if your current layout has the tabs `Teleoperated`, `Autonomous`, and `Testing`, and your layout defines a tab named `Testing`, after downloading, any content that you have on the `Testing` tab will no longer exist and will be overwritten by whatever is on the remote layout tab.

例如，如果当前布局有 `Teleoperated`、`Autonomous` 和 `Testing` 选项卡，而要下载的布局定义了一个名为 `Testing` 的选项卡，那么下载后，现有 `Testing` 选项卡上的全部内容都会消失，并被远程布局中该选项卡的内容覆盖。

Merge

合并

Merge the downloaded layout with the existing one. If a new widget cannot be properly placed, it will not be added.

将下载的布局与现有布局合并。如果新控件无法正确放置，则不会添加它。

If your current layout has the tabs `Teleoperated`, `Autonomous`, and `Testing`, and your layout defines a tab named `Testing`, but the layout you are downloading also has a tab named `Testing`, any widgets you currently have on the `Testing` tab will remain, but new widgets will be added in spots that are not occupied.

如果当前布局有 `Teleoperated`、`Autonomous` 和 `Testing` 选项卡，而且布局中定义了名为 `Testing` 的选项卡，同时要下载的布局也有一个名为 `Testing` 的选项卡，那么 `Testing` 选项卡中现有的控件会保留，新控件则会添加到未被占用的位置。

Full Reload

完全重新加载

Deletes the existing layout and loads the new one.

删除现有布局并加载新布局。

This is the same operation as opening a new layout locally, any widgets you have placed will be removed.

这与在本地打开新布局执行相同的操作，你已放置的所有控件都会被移除。

Custom Widget Examples / 自定义控件示例

Single Color View / 单色显示

```
Color exampleColor = new Color(68, 238, 255);
SmartDashboard.putString("Example Color", exampleColor.toHexString());
```

SwerveDrive

Note: Velocity must be in meters per second, and angles must be in either radians, degrees, or rotations, and CCW+

说明：速度必须以米每秒为单位；角度必须以弧度、度或转数表示，并且以逆时针为正（CCW+）

```
SmartDashboard.putData("Swerve Drive", new Sendable() {
    @Override
    public void initSendable(SendableBuilder builder) {
        builder.setSmartDashboardType("SwerveDrive");

        builder.addDoubleProperty("Front Left Angle", () -> frontLeftModule.getAngle().getRadians(), null);
        builder.addDoubleProperty("Front Left Velocity", () -> frontLeftModule.getVelocity(), null);

        builder.addDoubleProperty("Front Right Angle", () -> frontRightModule.getAngle().getRadians(), null);
        builder.addDoubleProperty("Front Right Velocity", () -> frontRightModule.getVelocity(), null);

        builder.addDoubleProperty("Back Left Angle", () -> backLeftModule.getAngle().getRadians(), null);
        builder.addDoubleProperty("Back Left Velocity", () -> backLeftModule.getVelocity(), null);

        builder.addDoubleProperty("Back Right Angle", () -> backRightModule.getAngle().getRadians(), null);
        builder.addDoubleProperty("Back Right Velocity", () -> backRightModule.getVelocity(), null);

        builder.addDoubleProperty("Robot Angle", () -> getRotation().getRadians(), null);
    }
});
```

YAGSL Swerve Drive / YAGSL swerve drive

If you are using the YAGSL Swerve Library, you can create a Swerve Drive widget without even modifying your code. Simply head to the SmartDashboard table, and drag the swerve table onto the grid, and you will have created a Swerve widget that shows your current and desired module states.

如果使用 YAGSL Swerve Library，甚至无需修改代码就能创建 Swerve Drive 控件。只需进入 SmartDashboard 表，将 swerve 表拖到网格上，就会创建一个 Swerve 控件，显示模块的当前状态和期望状态。

Alerts / 警报

This displays data from the NetworkAlerts by [Team 6328](#). For examples and implementation, see [Mechanical Advantage 2023 Robot Code](#)

此控件显示 [Team 6328](#) 的 NetworkAlerts 数据。示例和实现见 [Mechanical Advantage 2023 机器人代码](#)

For robot code using a WPILib version after 2024, this will be available in all WPILib projects and no additional files need to be added.

对于使用 2024 年之后版本的 WPILib 的机器人代码，所有 WPILib 项目都可使用此功能，无需添加其他文件。

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Privacy Policy / 隐私政策

Elastic is a volunteer-run project that is owned and published by Nadav from FIRST Robotics Competition Team 353 ("Elastic"), based in the United States.

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Information We Collect / 我们收集的信息

Elastic does not collect data from users of the application.

Elastic 不收集应用用户的数据。

The Elastic application will create and store logs of application errors and user interactions on the user's system, but these logs are not shared unless it's done manually by the user.

Elastic 应用会在用户系统上创建并存储应用错误和用户交互的日志，但除非用户手动分享，否则不会共享这些日志。

Third-Party Data Collection / 第三方数据收集

The Elastic repository is hosted by GitHub, which publishes a [privacy policy](#) that applies when interacting with these services.

Elastic 仓库托管在 GitHub 上。与这些服务交互时，适用 GitHub 发布的[隐私政策](#)。

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Contact / 联系

Questions about this policy can be directed to frc-elastic@outlook.com

有关本政策的问题可以发送至 frc-elastic@outlook.com

Shuffleboard API

WARNING

警告

Support for the Shuffleboard API has been removed in favor of downloading full layouts from the robot, see the [migration guide](#) for more information

Shuffleboard API 支持已被移除，改为从机器人下载完整布局。更多信息请参阅[迁移指南](#)

Elastic supports generating layouts with code via the Shuffleboard API. Most features of the API work the same way as they do with Shuffleboard. However, some features are either not supported, or behave in different ways. The documentation for how to use the Shuffleboard API can be found [here](#).

Elastic 支持通过 Shuffleboard API 使用代码生成布局。API 的大部分功能与在 Shuffleboard 中的工作方式相同。不过，某些功能不受支持，或行为有所不同。有关 Shuffleboard API 的使用文档见[此处](#)。

Tabs

选项卡

Tabs work the exact same way as with Shuffleboard, except the tabs created with code are not separated from tabs that are manually created.

选项卡与 Shuffleboard 中的工作方式完全相同，但通过代码创建的选项卡不会与手动创建的选项卡分开。

Setting Size, Position, & Widget Type

设置大小、位置和控制类型

Setting the size, position, and type of widget works the exact same way as it does with Shuffleboard. However, some widgets may not be implemented yet. To know if you can use a certain widget type, see [Widgets & Properties Reference](#). For the `widgetType` parameter in the `withWidget()` method, put the name of the widget exactly how it is displayed in the widgets reference. For example, if you wanted to add a ComboBox Chooser with the API, you would call `.withWidget("ComboBox Chooser")`.

设置控件大小、位置和类型的方式与 Shuffleboard 完全相同。不过，某些控件可能尚未实现。要确定某种控件类型是否可用，请参阅[控件与属性参考](#)。对于 `withWidget()` 方法中的 `widgetType` 参数，请准确填写控件参考中显示的控件名称。例如，要通过 API 添加 ComboBox Chooser，可以调用 `.withWidget("ComboBox Chooser")`。

Layouts

布局

Only List Layouts are supported.

仅支持列表布局。

Properties

属性

Properties are supported, however some parameter names may differ from widget to widget. To see the parameters for each widget, see [Widgets & Properties Reference](#). Be sure that the data types for each property are correct, otherwise widgets may not display properly.

支持属性，但不同控件的某些参数名称可能不同。要查看各控件的参数，请参阅[控件与属性参考](#)。请确保每个属性的数据类型正确，否则控件可能无法正常显示。

Recording

录制

Recording is not supported, and recordings will not be added. For logging data and viewing it after a match, we recommend using the [WPILib DataLogManager](#) to save logs to your roboRIO, and use a software such as [AdvantageScope](#) to visualize them. Recording logs directly from the roboRIO is much more reliable, will allow you to see data update much more frequently, and will not be affected by disconnections.

不支持录制，也不会添加录制功能。要记录数据并在比赛后查看，建议使用 [WPILib DataLogManager](#) 将日志保存到 roboRIO，并使用 [AdvantageScope](#) 等软件进行可视化。直接在 roboRIO 上记录日志可靠得多，能以更高频率查看数据更新，而且不会受断开连接的影响。

Shuffleboard API Migration / Shuffleboard API 迁移

WARNING

警告

This can only be done on the 2025 version of Elastic

此操作只能在 2025 版本的 Elastic 中进行

Code-Driven Layouts / 代码生成的布局

Migrating Shuffleboard API layouts to remote downloading layouts is simple:

将 Shuffleboard API 布局迁移为远程下载布局很简单：

1. Clear all tabs and widgets of your dashboard, it is recommended you save your layout to a safe location before doing this step

清空 Dashboard 的所有选项卡和控制件。建议在执行此步骤前，将布局保存到安全的位置

2. Connect to your robot to populate Elastic with the code-driven layout

连接到机器人，让 Elastic 加载通过代码生成的布局

3. Export your dashboard layout

导出 Dashboard 布局

4. Set up [remote layout downloading](#)

设置[远程下载布局](#)

After migrating your layout, it is advised to remove any Shuffleboard API-related code from your robot project.

迁移布局后，建议从机器人项目中移除所有与 Shuffleboard API 相关的代码。